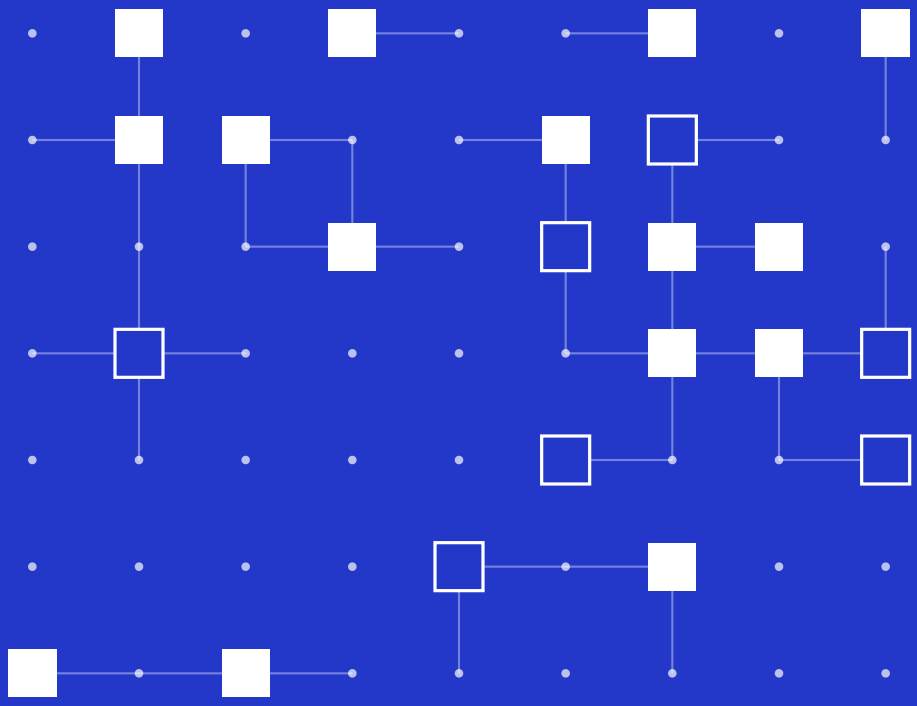
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programmable society

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§ preface

before we begin

this book was written in a month.

not planned as a book. it started as posts. one thought a day, sometimes five, typed between charts and calls and long walks, usually late at night when the timeline goes quiet and you can actually hear yourself think. october was a strange month to be in crypto. prices were loud, ideas were quiet, and almost everyone i knew was either exhausted, addicted, or both.

somewhere in the middle of that month i noticed that the posts were talking to each other. a thought about refundable tokens was really a thought about trust. a rant about airdrop farmers was really a question about contribution. a dream about satoshi was really a question about what we came here for in the first place. different days, different moods, one argument.

that argument is simple to say and hard to build: crypto was never supposed to be only a market. it was supposed to be a new way for strangers to coordinate around rules they choose, without asking

permission from someone sitting at the top. money was the first thing we made programmable. society is the next.

so this book moves in a line. first, what we set out to build and how we drifted. then what the casino is doing to us, inside our heads. then the war between anarchy and control that most people don't even know they are fighting. then the tools: programmable money, token standards most of the market ignores, new kinds of rules. then the missing social layer: identity, privacy, and reputation. and finally the thing i am most excited about, human commitment markets, where people put something at stake to become better rather than just to get richer.

i kept my voice. lowercase when i think fast, full sentences when i slow down, double dots when a thought keeps running. i am not a financial advisor. nothing here is a call to buy anything. i am just a curious explorer who has been in this space long enough to remember why it mattered, and stubborn enough to believe it still can.

if one chapter makes you build something, the month was worth it.

– hitesh.eth

a conversation with satoshi

Satoshi came into my dream last night.

He looked at me and said,

“What’s up, bro? Do you still have Bitcoin?”

I paused for a while.

My eyes got wet. I reached for some water, took a sip, and replied,

“Bro, I don’t have any Bitcoin left.”

He looked surprised.

“What happened? I thought you had plenty. You discovered Bitcoin before 99.9% of the people on the internet even knew it existed. You were so early. What went wrong?”

I stayed silent for a moment.

Then I said,

“Being early was a blessing that slowly turned into a punishment.

When I first discovered Bitcoin, it wasn't just about money. It was about freedom, curiosity, rebellion, and the possibility of a different world.

But somewhere along the way, everything changed.

People who came later wanted the Bitcoin we had. They wanted better entries, bigger bags, higher returns. And slowly, an entire environment was created where nobody ever felt they had enough.

If you had 1 Bitcoin, you wanted 10.

If you had 10, you wanted 100.

And if you had 100, you were terrified of losing them.

I got caught in that game too.

First, greed convinced me that whatever I had wasn't enough.

Then fear convinced me that whatever I had could disappear tomorrow.

And somewhere between those two emotions, I lost almost everything.

Not just Bitcoin.

I lost sleep. I lost peace. I lost years of my life chasing numbers on a screen.

The funny thing is, bro, I was never poor when I entered this game. I became poor the moment I started believing I didn't have enough.

And what happened between greed and fear is the answer to your question."

Satoshi looked at me.

And then he laughed.

I woke up.

For a moment, I genuinely felt like I had spoken to him.

I checked my wallet.

Not even five figures.

Then I opened CT.

People were still flexing their PnLs. Someone had made millions. Someone was calling the next 100x. Someone was angry about missing a trade. Someone who had already made life-changing money was desperately trying to make more.

And suddenly, I realised something.

Some of us entered crypto to become free, but ended up becoming prisoners of the very thing we believed would liberate us.

We spent years waiting for that one number.

The number after which we would finally stop.

Finally travel.

Finally spend time with our families.

Finally make music, read books, sit under trees, and live without constantly checking prices.

But every time we reached a number, the number changed.

The finish line kept moving.

And life kept passing.

I looked at my wallet again.

It's an amount that most people on CT wouldn't even consider retirement money.

But somehow, I feel retired.

Not because I made enough money.

Because I finally stopped needing more to feel enough.

I still love Bitcoin. I still believe in what brought me here. But I no longer want my happiness to depend on a green candle.

Maybe I lost the game everyone else is still trying to win.

Maybe I could have been a multimillionaire if I had made a few different decisions.

Maybe I will never recover what I once had.

And maybe that's okay.

Because for the first time in years, I don't wake up feeling that I am behind someone.

I don't need to win back my past.

I don't need to prove that I was early.

I don't need the market to validate my existence.

I have something today that I couldn't buy even when my portfolio was worth many times more.

The ability to enjoy an ordinary moment without wishing it were something else.

I thanked Satoshi.

I thanked God.

Then I picked up my kalimba and started playing.

A few minutes later, I lost myself in the music.

No charts. No PnL. No next 100x. No regrets.

Just a few notes, a little silence, and a man who finally had nothing left to chase.

Perhaps that was the freedom I came looking for in Bitcoin all those years ago.

I came to Bitcoin looking for financial freedom.

It took me years of losing money to realise that freedom begins when money stops owning your mind.

§ part 1

part one

the promise

what we came here to build, and how we drifted

every movement starts with a promise and ends up negotiating with it.

crypto's promise was not "number go up." it was that strangers could coordinate around transparent rules without trusted intermediaries. that ownership could be direct. that money could be private. that value could move between people without someone at the top skimming every flow and deciding who gets to participate.

this first part is about that promise and the slow drift away from it. not a story of villains, mostly. a story of convenience, liquidity, and a community that started measuring itself with the scoreboard of the system it wanted to replace.

§ chapter 01

1

the point was never price

i want to start with a word almost nobody uses anymore: curation.

The world needs a curation protocol and that could actually become one of the biggest usecase crypto can bring on the internet after payments.. and this space doesn't even talk about it.. and the same space used to talk the thing called TCR everywhere they go in conferences.. actual builders who care about real solutions are long gone.. and people like me who wants to talk about all these stuff rarely get an audience because algo push the content to audience who come from a different mindset.. it's very hard for a genuine crypto believer to stay here where all you see people talking price while the core objective of the space was never price discovery but building a decentralised society.

the idea we forgot

around 2017 the most exciting conversations at crypto conferences were not about memecoins. they were about mechanisms. token curated registries, TCRs, were one of them. the idea, popularised by

mike goldin, was beautiful in its simplicity: a list that nobody owns but everybody can trust. you stake tokens to propose an entry. others can challenge it by staking too. token holders vote. whoever was wrong loses their stake. the list stays honest because being dishonest costs money.

a list of trustworthy news sources. a list of legitimate projects. a list of good doctors, real merchants, safe contracts. the internet's biggest unsolved problem was never moving information, it was knowing which information deserves your attention. search engines and feeds solved that by becoming the curator, and the curator became the most powerful company in the room.

TCRs had real problems. voter apathy, plutocracy, the cost of challenging. they were early, and early is often another word for unfinished. but the instinct behind them was right: crypto could become the curation layer of the internet, not just its payment layer. and that instinct disappeared the moment price became the only thing worth discussing.

so let me state the thing plainly, because everything else in this book rests on it.

The whole point of crypto is to enable a set of rules through code and find the right coordination to enforce those rules in a community that shares some sort of common values to achieve a shared economy without needing third-party operators from traditional systems.

Now just read this again and try to analyze: are we really doing crypto the right way? If not, then that simply means there is scope for improvement. When there is scope for improvement, that simply means the introduction of new ideas, designs, and products, which are more likely to be fast-tracked using AI.

But since the industry has not been able to establish a security framework like OWASP, even if we progress from here, some of the hacks could put us way behind in terms of growth, and that's very concerning.

rules, coordination, and a shared economy

read that definition slowly. it has four parts: rules, code, coordination, and a community with shared values. price is not in it. tokens are not even in it. tokens are one way to express rules about value. they were never the point by themselves.

The original idea behind crypto was to empower more people, remove unnecessary middlemen, and return the value they were extracting directly to the parties transacting with each other.

Even when a platform was involved, the idea was to let the community decide its future rather than a few shareholders controlling everything.

and the honest question that follows is the one i keep asking.

what happened to the original promise of crypto??

we spent almost a decade talking about decentralization and financial freedom but how much progress have we actually made toward building systems where people can establish identity without surrendering privacy and build reputation without depending on centralized platforms and participate in economic opportunities without needing access to insiders..

we built some amazing financial infrastructure but somehow the conversation always returns to which coin is going to pump next..

we have created a massive speculative economy around technology which was supposed to change how humans coordinate with each other and i think we should seriously question whether we are building the future we once imagined or simply becoming better at trading the promise of that future.

the rest of this part is my attempt to answer it. not with nostalgia, but with a timeline, because the drift did not happen in one day. it happened through choices, and choices can be made differently.

§ chapter 02

2

two cultures

to understand where crypto is heading, you have to understand that in 2016 there were two very different groups saying the word "blockchain," and they meant opposite things by it.

back in 2016 there were two parallel cultures developing around blockchains..

the first culture grew around IT companies and banks exploring blockchain use cases. most of these experiments were happening on enterprise platforms like hyperledger fabric which had IBM as one of its major contributors. corporates were building their own permissioned chains and announcing partnerships and running POCs and trying to convince the world that blockchain would transform traditional finance.

but almost none of these experiments achieved anything close to the attention and scale public blockchains were getting.

the second culture was growing around bitcoin and ethereum. a culture built by cypherpunks and developers and early believers who were not trying to make existing institutions more efficient. they were trying to create an alternative to those institutions. the whole idea was to reduce our dependence on trusted intermediaries and give people the ability to own their assets and coordinate economic activities without needing permission from someone sitting at the top.

one culture wanted to make the existing system more efficient while the other wanted to make the existing system less necessary.

and that was the fundamental difference between these two cultures.

then something interesting happened.

public blockchains started making their technology available as public goods. cosmos played a major role in this transition by introducing an SDK that allowed anyone to build their own blockchain. other ecosystems eventually developed similar frameworks and suddenly building a blockchain became much easier than building the culture and community that made public blockchains valuable in the first place.

and this was the moment when the boundaries between these two cultures started disappearing.

the same institutions that once wanted to build their own enterprise blockchains began realizing that they didn't need to develop everything themselves. they could simply adopt the technology developed by public blockchain communities and create their own networks with different levels of permission and control.

they could borrow the architecture without adopting the philosophy.

they could use the same cryptographic tools without accepting the same political and economic commitments.

they could launch their own chains and issue their own assets and create their own ecosystems while retaining control over who participates and who validates and who ultimately benefits from the value being created.

and slowly the enterprise blockchain culture that failed to achieve meaningful adoption through its original experiments began finding a new identity inside the public blockchain ecosystem.

the irony here is that public blockchain communities spent years creating open infrastructure to reduce the power of centralized institutions and those same institutions eventually discovered that they could use this infrastructure to strengthen their own position.

and we welcomed them.

we started measuring progress through institutional adoption and the number of banks experimenting with tokenization and the amount of real world assets moving onchain and the number of stablecoins being issued.

we began celebrating the very institutions whose influence we once wanted to reduce.

and somewhere during this transition the definition of decentralization also started changing.

earlier decentralization was about distributing power and ownership and reducing dependency on centralized authorities.

now decentralization is increasingly being discussed as a technical property of infrastructure rather than a social and economic commitment.

a blockchain can have thousands of validators and still serve an economy where the issuance of money and access to assets and control over economic participation remain concentrated in the hands of a few institutions.

hyperdollarization

and this is where the whole conversation around hyperdollarization becomes important.

we spent years imagining a future where people could move beyond the monetary systems controlled by governments and central banks. bitcoin emerged from that imagination and ethereum expanded it by allowing people to build entirely new economic systems through code.

but look at what we have been celebrating over the past 3 years.

the expansion of dollar denominated stablecoins across public blockchains. the integration of traditional financial assets into defi. the increasing dependence of onchain economies on dollar liquidity and institutional capital.

we are building one of the most advanced financial settlement infrastructures in human history and a significant part of its economic activity is becoming another distribution network for the same monetary system we once wanted to challenge.

the dollar is becoming more programmable and more accessible and more globally distributed through the infrastructure developed by communities that originally wanted monetary alternatives.

and this is not necessarily a technological failure. stablecoins have created real utility for millions of people and public blockchains have made global value transfer more accessible.

but we need to distinguish technological adoption from ideological progress.

because they are not the same thing.

when a bank launches a blockchain it doesn't automatically become part of the cypherpunk movement.

when an institution tokenizes an asset it doesn't automatically distribute economic power.

when a centralized issuer launches a stablecoin on a decentralized network it doesn't automatically make the underlying monetary system decentralized.

and when the industry celebrates all these developments without questioning who controls the assets and who captures the value and who can censor transactions and who gets excluded from participation we are slowly turning decentralization into a marketing language.

the enterprise blockchain culture didn't necessarily defeat the public blockchain culture through better technology.

it learned to adopt its technology while preserving much of its original worldview.

and perhaps the most fascinating part is that many of us have started treating this adoption as the ultimate validation of everything we built.

we have reached a point where institutions no longer need to convince us that their version of blockchain is the future.

we are doing that job for them.

they are wearing the mask of crypto while continuing to serve the economic structures they were originally created to protect.

and the tragedy is not that institutions are adopting public blockchains. open infrastructure should be available to everyone.

the tragedy is that the people who built this infrastructure are increasingly forgetting why they wanted it to exist in the first place.

we wanted to create an alternative financial and economic system where individuals could exercise sovereignty over their assets and identity and participate in markets without asking permission.

we wanted to build sharing economies where value could move directly between people and communities could coordinate through transparent rules rather than institutional authority.

we wanted to make trust less dependent on powerful intermediaries.

but today much of our attention has shifted toward making those intermediaries more efficient.

and perhaps we should ask ourselves whether we are actually building a new economic civilization or simply upgrading the infrastructure of the old one.

because if the ultimate achievement of public blockchains is to make the dollar more dominant and traditional financial institutions more efficient then we need to ask what happened to the original promise of crypto.

just remind yourself who we were in 2016 and what we have become in the past 3 years.

> *one culture wanted to make the existing system more efficient while the other wanted to make the existing system less necessary.*

§ chapter 03

3

the matrix of the dollar

the previous chapter ended with a question about who we were in 2016. this chapter is about what the system we wanted to exit is doing with what we built.

do you remember that scene from the matrix where morpheus explains to neo that the world he believed was real was actually a system designed to keep people inside it?

i think crypto is approaching a similar moment..

we spent a decade building an alternative financial system to challenge the existing controllers of the economy.. and now imagine billions of people entering crypto only to hold dollars, trade tokenized american stocks and speculate on pre-ipo companies..

the more liquidity stablecoins attract, the more demand they could create for american debt.. the same infrastructure we built to challenge the existing financial system could eventually help it expand its dominance..

and the irony is that blockchain adoption could become extraordinarily successful while most native tokens capture very little economic value..

so what are we even building?

why this is not paranoia

let me add some mechanics here, because this argument often sounds like a conspiracy theory when it is actually just how balance sheets work.

a dollar stablecoin is a promise: one token, redeemable for one dollar. to keep that promise, issuers hold reserves, and a large part of those reserves sits in short-term US government debt. so every time someone in lagos or manila or thāne swaps their local savings into a stablecoin, somewhere a treasury bill gets bought. demand for onchain dollars becomes demand for american debt. the more successful stablecoins become, the more our open infrastructure works as a global distribution network for one country's monetary system.

that is not evil. for a person whose local currency is collapsing, a stablecoin can be a lifeline, and i will never mock that. but it is not the alternative monetary system we said we were building. it is the old system, upgraded with our tools.

and notice what happens to native assets in that world. if everything is priced, settled, and stored in dollars, then eth, btc, and every other native token becomes a speculative side bet sitting next to the real economy instead of being the money of that economy.

why can’t we create markets where people need eth, zcash, near and other assets to participate in productive economic activity?.. markets for labour, data, human commitments, reputation and ai

agents..

where people get rewarded for contributing something valuable rather than simply holding an asset and waiting for someone else to buy it at a higher price..

and where privacy, identity and reputation become infrastructure for economic freedom rather than another surveillance system..

the next crypto war is not about which token will outperform..

it's about who gets to build the markets..

because the matrix doesn't necessarily need to destroy crypto..

it could simply absorb everything we built and make us celebrate its expansion as our own victory.

private money needs an economy

if the matrix absorbs us through liquidity, the way out is not a better slogan. it is an economy where native money is actually needed. that is the lens through which i look at privacy coins, and it is why i am harder on zcash than most of my friends would like.

not many people wanted to hear it but the story of $ZEC as private money starts falling apart when you check its merchant adoption and the rate at which it has been growing.

the only positive metric has been its growing shielded pool which is a good thing but when you look at who could offer similar privacy with better network effects and bigger merchant adoption two possibilities come to mind. $ETH and stablecoins.

stablecoin payments will likely become increasingly private as privacy infrastructure evolves. more wallets will also make it easier to convert your coins into onchain dollars. the question is whether

we are building an alternative monetary system or simply becoming exit liquidity for the same dollar economy we once wanted to escape.

now if we really talk about cypherpunk money we can talk about $ETH and potentially every coin on ethereum once privacy is achieved across different layers.

shielding transactions alone isn't enough. you need transport layer privacy through tools like tor and protection against leakages from RPC providers wallets metadata and identity exposure.

only then can we talk about true private money.

and people will use it more when they realize stablecoins are becoming the new version of the same matrix we wanted to exit when we discovered bitcoin.

we almost killed the BTC as money narrative and accepted stablecoins as the new normal because they offer more liquidity. but that wasn't the case during the 2016–17 cycle when BTC dominated crypto liquidity.

we moved from building an alternative monetary economy to building better rails for the dollar economy while celebrating it as crypto adoption.

now if you want to create real liquidity around private money you need to build an entire ecosystem around it. an economy where people create earn exchange and share value in a less extractive environment.

this is where identity reputation and programmable commitments become important.

imagine an economy where humans and AI agents exchange skills services knowledge and products using private money. where identity can be verified without exposing personal information. where reputation is earned through contributions and commitments are enforced through code rather than centralized authorities.

where rewards come after real work rather than speculation.

this creates actual reasons to earn hold and spend money within the ecosystem instead of constantly converting everything into dollars.

and i think ethereum has the strongest foundation to attempt this because of its liquidity programmable infrastructure developer ecosystem and growing zero-knowledge capabilities.

privacy alone won't create monetary adoption. you need economic activity around it.

we have spent years designing tokens people can speculate on while barely designing economies where people actually need those tokens.

the future of private money won't be decided only by who offers the strongest privacy. it will be decided by who builds the strongest economy around that privacy.

and that's where i think ethereum has a much bigger opportunity than most people realize.

the intruders

i will say this one with less diplomacy, because i felt it with less diplomacy.

If our collective understanding of crypto and blockchains remains as low as it is today, then the day is not far when TradFi will reintroduce all those blockchain use cases which we once explored in 2017, and make them sound like something that could benefit us.

They are more like intruders who stole our ideas, halted our growth, and are now accelerating their dominance.

It's more like we were all in a war against them. Many of us got killed when our capital got wiped out because, in this war, we were backing our beliefs with capital.

And now, when we have the perfect playground to start the war again, people who are supposed to be on our team are actually celebrating the achievements of the kingdom we started the war against.

Just try to understand what we are even doing.

§ chapter 04

4

the forgotten

every industry tells a story about who built it. crypto's current story is told mostly by people who arrived after the hard part.

whenever we talk about the growth of crypto or defi, people from the web3 workforce often try to create an image that they brought this growth by putting out some numbers, but the real heroes are those who sacrificed both capital and time when most of the world didn't even believe crypto would survive.

those founders from the 2016-17 era who tried to build great ideas with limited resources are now forgotten like their efforts never mattered.

why can't we invite those OGs to the events you are hosting? you don't because they can't sponsor you.

you think their opinions don't matter anymore, but the deeper connection they shared with crypto can't be matched by most founders building today.

they genuinely believed in decentralization, ownership and a sharing economy. some were simply too early to succeed, and being too early became their punishment rather than their achievement.

most of them eventually quit crypto, took other jobs, started something else, and some probably never recovered from what they sacrificed. imagine spending the most productive years of your life building something the world refuses to understand, only to watch the same industry become a trillion dollar economy where people who arrived much later are celebrated as visionaries while nobody remembers your name.

and if those people think about returning today, even a single day on CT might disappoint them. the space they once considered home has been hijacked by casinos and fintech institutions.

freedom has been replaced by liquidity, ownership by speculation, and the sharing economy by extraction. it looks like a dead sea, full of capital but losing the soul that once made this movement meaningful.

but there are still people trying to remain delusionally optimistic about the good old days of crypto, when unity, shared values and a decentralized future mattered more than quick gains.

maybe they are carrying something this industry desperately needs, a memory of why we started this journey. we have become so obsessed with measuring growth through numbers that we have forgotten how to measure the loss of values, trust and people who built the foundation.

maybe the next chapter of crypto should begin by listening to those forgotten people again.

an industry that forgets those who sacrificed everything to build its foundation shouldn't be surprised when the next generation arrives only to extract everything it can.

the network that ran on goodwill

there is an older lesson that makes this even sharper.

BitTorrent actually proved you don't need to drive incentive to create a network with deeper effect, and million of users everyday. That was a real crypto product which purely runs on a human helping other human because they both sharing a common passion for good content. It's too hard to think about building BitTorrent again without incentive. Such audience which care more about cause than money is gone. And people got so habitual to see some of incentives if they are in crypto than they can't even digest if someone say do it for free because this is something you care about.

bittorrent did have one clever incentive. bram cohen designed it so peers who upload get faster downloads, a tit-for-tat rule baked into the protocol. but that rule only shaped behaviour inside a swarm. it did not explain why millions of people kept seeding files for strangers long after their own downloads finished. that part was culture. people shared because sharing made the network better, and the network was theirs.

crypto was built by people who were willing to do the impossible for free.. now we struggle to get people to do anything without paying them..

trying out something which looks impossible to achieve at the beginning, and slowly getting closer to making it possible, gives you more joy than making millions of dollars..

the bitcoin network grew on the same enthusiasm in its early days.. people contributed their computing power to mine bitcoin, not because they knew it would become valuable someday, but because they believed they were participating in something extraordinary.. financial incentives existed from day one, but imagining bitcoin reaching today's scale was almost an impossible dream.. and that dream was backed by curiosity, experimentation, ideological conviction and altruism before capital entered and changed the whole dynamics..

and bitcoin wasn't even the beginning of this culture..

study how bittorrent worked.. millions of people shared their bandwidth and computing resources to distribute files across the internet.. there was no native token, no speculative market, no promise of becoming rich.. there were practical incentives like reciprocity, but people also contributed because sharing made the network stronger..

this was an extraordinary example of a voluntary digital society.. strangers coordinating with strangers, sharing resources and collectively building infrastructure without depending on a central authority..

and i think creating such voluntary digital societies remains one of the most beautiful promises of cryptography and decentralized networks..

the sad part is that we barely discuss this anymore.. we attend hundreds of crypto events and conferences talking about liquidity, valuations, token prices and institutional adoption.. but rarely ask how we can inspire millions of people to voluntarily contribute toward something that looks impossible today..

perhaps we need to rediscover the joy of building something together before asking how much money we can make from it.

> *perhaps we need to rediscover the joy of building something together before asking how much money we can make from it.*

§ part 2

part two

the casino inside us

what the market is doing to our minds

before we talk about mechanisms and token standards, we have to talk about people. because no protocol design survives a community that is burned out, addicted, and comparing itself to the luckiest person on the timeline every hour.

this part is uncomfortable. it is about the game we play inside our own heads. i am writing it as someone who has played that game, lost at it, and slowly learned to step outside of it.

§ chapter 05

5

ready player one without a headset

do people still cover projects the way they used to cover with all details

Also since now everyone can actually do research on anything they want so is it even relevant for a new creator to breaking down tech and projects.

Or we can simply say all they want tickers now nothing else because words are overwhelming for many all they need a key to become rich.

So in that context don't you think we are in a game like ready player one already and you don't even need VR headset to feel immersion.

You are immersed fully into this reality which dominate your mind body and heart.

And when you realise that how hard is it to leave the house even when you know you get no parents in the house .

It's just people fighting each other PvP.

And you are fighting too but the problems here is bread is even before you get it to public all you get breadcrumbs and the fight for getting the max share of it but that share is even sizeable when you look outside the house called CT so it makes sense to stay and keep fighting

What a life lol

the mood follows the chart

wake up check chart feel good or bad the whole day..

and when you deal in crypto markets where you get few good days and more bad days.

You eventually lose mood to do anything new.

You surrender yourself to procrastination.

You just grow self doubt if you don't win enough or you just grow self confidence if you win enough both are harmful in the end when market stops giving you high even for few weeks.

Now the question is : is this a good life or we can do better?

this is the part nobody puts in the pitch deck. markets give you a few good days and many bad ones, and if your mood is tied to the chart, your life becomes a function of something you do not control. winning makes you overconfident. losing makes you doubt everything. both are dangerous when the music stops.

"Good times" are very limited when you are in crypto.. you spend most of your time wishing, praying, crying, sobbing.. thinking of quitting, but it's too hard to quit.. it's an addiction, but again, once you are free from the addiction loop of crypto markets.. and then you begin to look at crypto again.. your whole lens would change, and the same crypto which looked like a bad deal to you becomes

the best thing in the world, where you feel like, how can I contribute to building something, not to trade but to do something real, where I could become part of the whole journey and have more control over my choices.. that's when you develop an entrepreneurial mindset, and you play around and figure out something.. you might fail many times while doing those projects, but it's better to fail as a founder than as a trader because you gain more knowledge, networks, and relationships which help you reach a good position in a few years, even when all your plans have failed hard.

living ten years ahead

there is another strange side effect of being deep inside this world for too long.

we are living different lives under different laws and when you live inside ct for a longer time considering your feed doesn't serve you thirst traps or greed traps but rather advancements in tech

and their real life implementation and effects then you actually grow a bigger disconnect with people who are technically 10 years behind in terms of vision and mindset and they are also operating at a very slow conscious processing rate with very bad cognitive health.

you can't even develop a healthy dialogue with them. either you or they will be pissed if you spend over an hour together where you are free with your expression. you will simply dominate and they will fear submitting to your ideas so either they will become toxic or label you out. no other way.

and the real tragedy here is that such phenomena are happening even within a very small internet collective society called ct where people come from different schools of thought and experiences and

cultures and they can't even connect with each other many times often creating huge information asymmetry gaps which are exploited by extractors while we remain hopeful for the next 100x.

attention is the scarce resource

and the cycle is only getting louder.

In this age of abundance, the most scarce resource on earth is no longer information - it is focused attention.

In context to crypto cycle.. you will actually see 100 more interesting things to play around

Because now we have more devs

More ideas will be realised into products

And the fragmentation of the cycle would be 10x more than what you saw in the past.

You got options: identify 5 projects build position and show patience.

Or you just move attention but make sure you pick a niche and move attention within that.

Just master a niche it could be privacy, Perps, prediction, new token designs, hooks or even launchpads .

§ chapter 06

6

conviction, information, and judgment

if the last chapter was about emotion, this one is about decision making. because a lot of what we call "alpha" is really a story about psychology wearing a data costume.

the same information, different lives

Imagine there is an opportunity in the market where everyone knows that a particular coin might pump by 2x and we have so many coins like this.. hint: projects which are solving identity, reputation. (Proven ones not larps)

Now Everyone has access to the same information.

Will everyone make the same decision?

Of course not.

Some will buy immediately, some will wait, some will procrastinate, some will avoid the risk, and some will simply ignore the opportunity.

Because information is only one part of decision-making.

The other part is human psychology.

Your habits, your past experiences, your willingness to sacrifice, your financial condition, and your ability to act determine what you do with the information.

This is why I don't think information symmetry should be confused with equality of outcomes.

The objective should be to give people a fair opportunity to access information.

What they do afterward will always be different.

We don't need everyone to make the same decision. We need people to have the opportunity to make their own decisions without someone deliberately keeping them ignorant.

when conviction becomes a trap

bitcoin created a culture where people started believing that holding something for a longer period could eventually change their financial circumstances..

it worked for bitcoin and then it worked for ethereum and ripple and many other early coins..

people were not just investing their capital they were investing their emotions too and that's how the whole diamond hands culture became popular..

but then people started applying the same philosophy to every coin without really understanding what they were holding..

the same conviction which once helped people create wealth eventually became the reason why many people couldn't exit from assets which had no meaningful economic foundation..

conviction without understanding is not a strength it can become the biggest weakness of an investor..

why i don't shill tokens

Even if you do deep research and share a detailed thesis with clear disclosures, you can't avoid some of your bets failing, and that's when some people will make you their scapegoat.

They won't even read and understand the whole post because they just operate on a pre-existing bias, and all they look for is confirmation.

And when they come across someone with a reputation for good calls, they will follow you blindly, and when they get trapped, they grow negative emotions and pass them on to you too.

I realized this deeply in the last cycle, and that's the reason why I don't talk about tokens like everyone else does on this platform.

Somehow, I feel talking about tokens eventually brings more negative emotions in the long term.

And since I am not interested in pleasing anybody or growing my numbers, I won't take that route unless I find something where I could build conviction myself.

At this stage, I don't really have any conviction in any token, and neither am I that rich that I can buy many coins and just talk about them, hoping I will make some good exits.

The only thing that excites me right now is thinking about the future and what role crypto is gonna play in shaping it.

Gonna have more mental exploration around it and share my findings and even projects that are doing something in that context, but make sure you understand that I don't really have any intention of talking about tokens.

So never take it otherwise.

I am not a financial advisor.

I am just a curious explorer.

the number that keeps moving

i want to end this part with the moment that made me rethink everything about incentives in this space.

sometimes i think what was the fault of people who were incentivized to do certain tasks? it actually changed millions of lives. gave them extra bandwidth to buy a gift for their loved ones. pay a bill.

breathe a little..it gave them hope. they felt CT was their home. some had no education. many never learned any other skill. all they thought was the market would keep paying. airdrops would keep happening. and everything around them made them believe that.

but even when airdrops were happening they felt sad after checking what others made. imagine getting $100 and feeling happy because you could finally buy something for someone you love. then seeing a random guy make $10k. your $100 is still there but the happiness is gone. nothing changed in your wallet but something changed in your head.

this whole habit of flexing how much you made and how much you got is dangerous for the space. someone reads your PnL as a judgment on their own life. they feel behind. they take bigger risks.

they start chasing a score they weren't even thinking about before opening this app. i wish someone would find a way to keep these PnLs hidden from the public forever. but will @fomo do it? i don't think so. it would hurt their business. a player can choose not to flex their score but will they? no..so the flexing keeps happening and we keep feeding the same loop.

just reflect for a moment. how many times were you happy with what you made until you saw what someone else made? we are stuck in multiple such loops in CT. and if we can see something is wrong then we should try to do something. and if i'm someone you look up to then i should give you something to raise your hope again. something where you can learn again. something like learn to earn. where you're learning and earning.

that is the loop. same wallet, same hundred dollars, different feeling. nothing changed on the chain. everything changed in the head. if we want to build better economies, we have to design for that head, not just for the wallet. hold onto this thought. it comes back in the last part of the book, when we talk about markets where what you learn stays with you even after the incentives stop.

§ part 3

part three

anarchy and the second crypto war

the fight most people don't know they're in

"anarchy" is a word that makes people nervous. it shouldn't. in this book it does not mean chaos. it means the absence of rulers who can change the rules to benefit themselves.

crypto sits between two forces. one wants open, permissionless, private coordination. the other wants the same technology with a master key. this part maps that fight, separates good anarchy from bad anarchy, and argues that without a real philosophy, every movement eventually gets reduced to price.

§ chapter 07

7

the second crypto war

the first war

most people in crypto today don't know there was a first crypto war, so let me tell it briefly.

in the 1970s, public key cryptography arrived and for the first time ordinary people could, in principle, protect their communications with mathematics strong enough to resist governments. governments did not like that. through the 80s and 90s, strong encryption in the united states was regulated like a weapon under export controls. when phil zimmermann released PGP in 1991 so anyone could encrypt email, he spent years under criminal investigation. in 1993 the US government proposed the clipper chip, an encryption chip with a built-in backdoor for authorities. cypherpunks fought back with code, papers, lawsuits like bernstein v. united states, and by printing source code in books because books were protected speech.

by around 2000 the export controls were mostly relaxed and strong encryption became the quiet foundation of the internet. every https padlock you see is a trophy from that war. bitcoin came out of that same culture.

the second war

crypto is divided right now.

west side: crypto anarchy (memes, nfts, and we need a casino with 100+ new possible markets)

also east side is crypto authoritarian state (tradFi chains, stablecoins, and anything and everything which comes under government regulations)

and we're in the second crypto war.

the first crypto war happened in between 1970-2000 almost for 28 years.

we entered into the second war this year, and we cant afford to wait more, and we need to accelerate crypto anarchy movement now if we really care about creating fair opportunities for many who're suffering from extraction from centralised systems and authorities.

but before that we need to kick out intruders from the house who're posing like they're with us but they're not.

the definition of crypto anarchy is very simple.

a liquid market for anything and everything you can put into words and images.

just give some thought to this statement, any wise guy would agree that ..we're growing crypto anarchy.

but also take the other side into account which is more like the opposite of anarchy: the crypto authoritarian side.

tradfi 2.0, hyperdollarization, RWAs.

the tech is the same.. public blockchains.

the design is the same.. creating an immutable, self-updating ledger which you could trace back to the genesis block.

it's just the intention that's different.

it's just that the people who are producing blocks are from the same company, not distributed like what we saw with bitcoin and blockchain.

so the kind of blockchains corporates like are less distributed, less decentralised, and more controlled, where they could even trace your money movements better than before. they can also turn off the blockchain anytime they want.

CT of 2017 would have revolted against these new blockchains already, like when we used to troll enterprise blockchains pretty hard.

they had no chance to stand in front of us, and now the same folks who once raised funds from our people through ICOs are operating at their command, wearing suits, promoting them, completely forgetting the fact that if people hadn't committed funds to their ICOs, they could never have built the businesses they're proud of.

the backdoor in the playground

i saw the same pattern inside the markets too, in a month where memes, tokenized stocks, and dollar rails started trading on the same screens.

What you have seen in crypto markets in past 3 months are extension of two sides: anarchy and authoritarian state.

When you make a meme pairing against tokenised stocks the whole objective there is to grow authoritarian dominance by creating a backdoor in crypto anarchy.

When you try to make the crypto anarchy playground less extractive by any tools or features you are simply creating an environment where more people join the anarchy.

But as long people are joining for anarchy eventually off ramping to stocks or fiat the whole purpose of anarchy fails right away.

So to achieve true crypto anarchy we need to figure out token ownership models, new token designs to make the playground more fair for newcomers where they could have higher chance to make gains.

Crypto anarchy for now is centered around making money, and as long the money is being made by participating in sharing economy it makes sense

But means of making money in anarchy is limited to token trades the environment should be created in a way where people get different options to make money .

When you completely change the definition of money in those environments where even your simple meme becomes more like money because there is an ecosystem around which supports exchange for goods or even cash while being complete private than it all makes sense in context to anarchy.

So to achieve anarchy we need to enable wider confidential txs, private swaps, new token designs, more ways to make the casino less extractive and a culture which have deeper philosophy embedded.

Without the philosophy we can't really stand against authoritarian states.

We need feel, breathe and understand what you are doing is an act of crypto anarchy it's just people are few years late to realise it.

§ chapter 08

8

good anarchy, bad anarchy

if the last chapter drew the line between anarchy and control, this chapter draws a second line, inside anarchy itself.

I have been thinking about anarchy in the context of crypto for a long time.

And I think there is a thin line between good anarchy and bad anarchy.

Good anarchy allows people to coordinate freely without depending unnecessarily on centralized authorities.

Bad anarchy emerges when people use that freedom to exploit others while avoiding accountability.

Think about crypto markets.

We have created systems where anyone can launch a token, create a market, and attract speculation.

That freedom is extraordinary.

But when the same freedom becomes an opportunity to manufacture narratives, manipulate attention, and extract money from uninformed participants, we should question the incentives.

The answer is not necessarily more centralized control.

Perhaps the answer is better identity, reputation, and enforceable commitments.

Freedom without accountability can become exploitation.

Accountability without freedom can become authoritarian control.

The future of crypto should be about finding a better balance between the two.

tokenizing belief

internet was open and then they censored it. they began gatekeeping information and crypto emerged as a tool of resistance against those who controlled the internet.

i mean the surface internet because the dark web which looks bad is actually an ocean of information and value exchange in its most naked form without the moral obligations of society.

true anarchy exists on the dark web where a good cause eventually became a bad habit. the same behaviour can be observed in another growing area of crypto -> markets.

tokenizing belief is the game.

you just believe in something for a second and we will create a market for you.

@Pumpfun is anarchy but it's bad anarchy. the moment you plug moral commitment hooks into it it could become good anarchy.

there is a thin line. helping others is good anarchy. knowingly extracting from others is bad anarchy.

the relationship between many KOLs and VCs and retail participants explains this perfectly.

so we need good anarchy and for that we need identity and reputation.

two of crypto's biggest bottleneck problems.

anarchy doesn't mean the absence of rules. it means the absence of rulers who can change rules to benefit themselves. crypto was supposed to enable coordination around shared values through code without centralized authorities.

but we made one fundamental mistake. we created financial identities before human identities. we gave everyone a wallet but never built a meaningful way to measure what value they contributed.

imagine a crypto economy where your identity doesn't reveal who you are but your reputation reveals what you have done.

you helped someone learn a skill you earned reputation. you built something useful you earned reputation. you fulfilled a commitment you earned reputation. you knowingly exploited others your reputation carries the consequences.

zk proofs could help verify certain contributions without exposing personal information while allowing reputation to travel across communities without compromising privacy.

but reputation alone isn't enough. we need moral commitment hooks embedded into economic coordination.

imagine every tokenized belief comes with commitments from those creating and promoting its market. those commitments could be enforced through staking and slashing and transparent disclosures and verifiable outcomes.

not every moral obligation can be translated into code but those that can should become part of the economic design.

your freedom to create markets remains intact but knowingly exploiting others comes with consequences.

we don't need another authority deciding morality for everyone.

we need communities defining shared values and enforcing commitments through transparent rules.

identity establishes continuity. reputation establishes trust. commitments establish accountability.

together they could move crypto from tokenizing beliefs to tokenizing meaningful human coordination.

the real revolution of crypto was never about giving everyone the freedom to create money out of thin air. it was about giving everyone the freedom to create value without permission while protecting those who create it from those who merely extract it.

> *identity establishes continuity. reputation establishes trust. commitments establish accountability.*

those three sentences are maybe the shortest version of this whole book. we will spend a full part on identity, privacy, and reputation, and another on commitments. but first, the tools.

§ chapter 09

9

a philosophy for the outlaw code

there is one more thing crypto anarchy needs before it needs better tokens. it needs to know what it believes.

Crypto Anarchy Needs an Actual Philosophy

There is another reason all of this matters.

Crypto sits between two forces: anarchy and authoritarianism.

Every centralized bridge back into traditional finance increases the influence of the authoritarian side. Every private transaction system, permissionless protocol, and autonomous economy strengthens the anarchic side.

But crypto anarchy cannot simply mean: buy token, pump token, sell token, withdraw to fiat.

If everyone eventually returns to the same financial system they claimed to escape, then the experiment never really succeeded.

Crypto needs an internal economy. More goods exchanged for tokens. More services. More labour. More private payments. More coordination. More ways to earn. More ways to participate. More ways to create value without simply trading one token against another.

And most importantly, it needs a deeper philosophy.

Because without philosophy, every movement eventually gets reduced to price.

code, culture, and belief

It's a high time to think about that why crypto has not became the cultural symbol against governments yet.. it got everything to become much bigger symbol than what an anime series called one piece is becoming.. also why we haven't do anything to capture the biggest youth distribution from largest economy who is showing resistance and shared the actual meaning of crypto to them while creating a sharing economy for them where those products could get distribution which are focusing on self custody of assets, privacy of assets, and creating verifiable identity for humans..I know you will be like we don't have clear regulations but did those exchanges waited for it when they turned crypto into a casino while not being able to secure their customers too. We need to rethink about our choices if we really care about crypto.

Understanding Religion is very very important in the age of AI not from the stories our culture told us but from the experience we personally had dealing with religions. Also we need to figure out scientific explanation to all those different terms described in ancient books, finding meaning through study becomes more accessible and easy to everyone. You don't need to find a guru for that. Most of them guru in India can't even take your 10 self enquiry

question they are all fake. Evoke your inner guru. Crypto is no different than religion. It gives you right to live, it creates common good, it promotes equality, it promotes freedom and when I say I mean cryptology here. The tech which makes crypto markets possible which is liberating in it all sense considering they could give you right to make the money private (that's a bigger unlock) because religion and money are so connected now and AI gonna play define the religion of the future and when you talk about agentic society, the value gonna shared in tokens. So talking tokens from philosophical lens becomes an essential requirement to my understanding. I will keep doing my bit to that direction

the algorithm problem

Algos are the atomic bomb of this generation.. efficient enough that they sent millions of youth thinking the largest democracy is in the danger.. who controls these algos?.. America.. these algos are like energy centers of millions of agents began to work together in this world already.. and that number would go to trillions and more than that too.. more smarter beings, threat to humanity, the center is sits right there in the last lender or resort we call it fiat currency it's sad new that crypto is being used to make that fiat currency more popular and people from crypto doesn't even care about it.. well this post gonna be wasted too but I will keep speaking

i wrote that one angry. let me say it calmer. recommendation algorithms decide what billions of people see, which means they quietly decide what billions of people believe is urgent. the companies running them sit inside one jurisdiction and one monetary system. if crypto's biggest cultural achievement ends up being "made the dollar easier to hold," then the same centre gets stronger on both sides: what we think and what we use as money. that is why i keep speaking, even when the posts feel wasted.

§ part 4

part four

programmable money

token design is social design

this is the part of the book where the philosophy turns into tools.

most people meet crypto through one kind of token: fixed supply, freely transferable, priced by a market, held in hope. but a token is just a rule about value, written in code. and rules can say almost anything. money can have a purpose. a token can have an undo button. a balance can expire. a membership can be portable. an nft can be a character, a comic, and a song at the same time.

ethereum's ERC process has produced hundreds of these ideas, written by people who thought hard about them, and the market has ignored most of them because everyone was busy launching the same token design again and again. the next chapters walk through some of my favourites, with the mechanics, the catches, and the businesses they could become.

§ chapter 10

10

words become rules

The whole point of crypto is actually simple: you create rules through code, then people coordinate around those rules.

A community shares some common values, and an economy starts operating without depending on the same third-party operators traditional systems require.

Once you understand crypto from this lens, you stop seeing tokens as things that only go up and down in price. Tokens become **rules around value**.

And that is where token design becomes interesting again.

A token does not need to be a simple transferable unit with a fixed supply. It can carry rules. It can represent rights. It can define conditions. It can change behaviour based on time, ownership, identity, reputation, or other onchain conditions.

Once money becomes programmable, token design becomes much closer to social design.

from kings to code

Kings used to define the rules of money. The kingdom enforced those rules.

Now code can enforce them.

And increasingly, whoever writes the code defines the rules.

The strange part is that AI will probably write more and more of that code. Give AI agents onchain identities, wallets, permission systems, reputation, and programmable assets, and you start getting something that looks like an alien economic society operating alongside humans.

Agents could enforce rules around money. Humans could then choose which systems make sense to them. Crowds would coordinate around certain forms of money, and those crowds would still need stories.

Stories need words. Increasingly, those words will become prompts.

So somewhere down the chain, you get something like this: **words become prompts, prompts become code, code becomes rules, and rules define money.**

i first wrote this idea in a rawer, angrier form, and i still think the raw version carries something the clean one loses.

Any and every token could become private money in next 2 years on ethereum natively.

Once you give agents an Onchain identity, you have an alien society enforcing rules of money for humans.

Now humans can decide which money looks more logical to them based on where the crowd agrees more, the crowd needs a group of herders, herders need stories, stories will use words.

So in the end words will enforce rules around money, and those words are simply prompts which would be given to agents by humans as long they have agency over them once the agency is snatched away due to rough behaviors agents themselves will decide enforce implement money in society but since the tribalism will grow a lot money as logic would have different faces and we all gonna see a Intial display of it very soon in crypto in next 3 months.

the chain

let me make that chain explicit, because it is the backbone of this part.

→ words become prompts.

→ prompts become code.

→ code becomes rules.

→ rules define money.

→ and money shapes how a society behaves.

whoever writes the words at the top of that chain has enormous power. today that is a small number of developers and institutions. tomorrow, more and more of that code will be written by AI, prompted by whoever has agency over it. that is why i think token design literacy matters for everyone, not just engineers. if you can read the rules, you can choose which ones to live under.

That sounds philosophical, but I think we are much closer to it than people realize.

§ chapter 11

11

money with a purpose

let's start with the simplest programmable money there is: money that knows what it is for.

onchain gift vouchers.

fund a smart contract with 10,000 USDC and issue 1,000 voucher tokens.

each token represents 10 USDC that can be redeemed at a participating online store, offline store, or event.

thats possible.

its called purpose bound money.

you define the purpose.

you define where the money can be spent.

and you define the conditions for spending it.

for example, you could build:

→ food vouchers redeemable at approved grocery stores.

→ education grants redeemable with approved course providers.

→ medical aid redeemable at participating pharmacies.

→ event credits for food, merchandise, or tickets.

→ employee allowances for meals or travel.

here is the interesting part.

the customer spends the voucher.

the contract checks the conditions.

the redeemed voucher gets burned.

and the merchant receives the underlying USDC.

the merchant doesnt have to keep circulating your voucher token. they receive payment.

you can also build rules around expiry dates, spending limits, and whether vouchers can be transferred to someone else.

and you could apply these rules to an AI agent's spending wallet.

give an agent a budget for approved services, and have the contract check its payments against that budget and the allowed recipients.

the agent chooses what to buy within the permissions you give it.

ERC-7291 provides a standard for building this using an ERC-1155 voucher backed by ERC-20-compatible money.

but the real-world connections still matter.

approving a supermarket doesnt automatically prove someone bought food. item-level restrictions need checkout integration.

releasing a grant after a milestone needs a reliable way to verify that milestone.

and users should know when vouchers expire, who can revoke them, and what happens to unused funds.

great token standard for distributing grants, aid, charity, and promotional budgets with specific spending conditions.

you can track where vouchers were redeemed onchain.

verifying what that spending achieved is the next layer.

this is not just a crypto idea

it's worth knowing that central banks are exploring the same concept. the monetary authority of singapore, through its project orchid work, published research on "purpose bound money" in 2022 and 2023, describing exactly this pattern: a digital token wrapper that carries spending conditions, sitting on top of an underlying money that the merchant ultimately receives.

that should tell you two things. first, the idea is serious enough that regulators are designing around it. second, the question is not whether programmable money will exist, but who gets to write the purposes. a government can use purpose bound money to deliver aid efficiently. it can also use it to restrict what citizens are allowed to buy. the same rule engine serves both. that is why i want communities, charities, brands, and individuals to be fluent in these tools too, not only states.

12

an undo button

crypto almost never gives people a way back. you buy, the market moves, and you live with it. this chapter is about what happens when a token is designed with an exit right from day one.

Tokens Don't Need to Behave Like Normal Money

This is where token standards become interesting.

What if money could expire? What if money could only be spent for a specific purpose? What if a token could be refunded? What if holding it created a different right than trading it? What if tokens had privacy by default? What if money behaved differently depending on the community using it?

We already have token standards exploring many of these ideas. The problem is that the market barely looks at them because everyone is busy launching the same token design again and again.

refundable tokens with an expiry date could be great for the ecosystem.

> you launch a token.

> people trade it.

> but until a fixed deadline, holders can return it for a predetermined refund.

> a token with an undo button.

its possible on ETHEREUM.

so “ONLY POSSIBLE ON SOLANA” is not a serious argument for this kind of token design.

Solana has documented token extensions and custom transfer hooks too. but Ethereum also has token standards with defined interfaces, documented behavior, and security considerations that builders can study and extend.

look at these two:

ERC-5507 introduces refundable tokens.

ERC-7818 introduces expirable token balances.

now understand the difference.

a refund right expiring is different from the token itself expiring.

for a tradable coin, you could let the refund right expire while the token continues trading.

example:

launch price: 1 USDC.

refund value: 0.80 USDC.

refund window: 7 days from launch.

during those seven days, holders can either sell at the market price or return their tokens for 0.80 USDC each.

the returned tokens are burned.

after the deadline, refunds stop. trading continues.

if someone buys at 2 USDC, their refund is still 0.80 USDC. it is a fixed redemption amount, not protection against every buyer's losses.

and transferring the token does not restart the countdown.

but the refund needs real backing.

if 100,000 outstanding tokens are refundable for 0.80 USDC each, you need 80,000 USDC reserved for those obligations.

that includes any team or liquidity tokens carrying the same right.

you cannot count the same money as refund backing, trading liquidity, and the team budget.

a hook cannot refund money that does not exist.

here is a starting prompt for the custom implementation:

"Build a Uniswap v4 Refund Window Hook with a companion token contract and refund vault. Let the token trade normally while its current holder can redeem it for a fixed amount until an immutable deadline. Fully reserve every refundable token before issuance. Burn tokens on redemption and pay the holder atomically. Keep refunds directly accessible without relying on the pool. Transfers must not reset the deadline. After expiry, trading and liquidity withdrawals must remain available. Publish who receives unused reserves. Build tests for reserve solvency, double refunds, and expiry boundaries."

Uniswap v4 provides swap and liquidity callbacks plus custom accounting, but the hook is only one part of this system.

study ERC-5507 for refunds. its ERC-20 interface specifies native ETH refunds, so a USDC implementation needs an explicit adaptation.

study ERC-7818 if you also want actual token balances to expire. it is not required just to put a deadline on refunds.

and understand liquidity pool risks before you launch.

if token balances inside a pool expire, pool accounting and withdrawals can be affected. ERC-7818 explicitly discusses liquidity-pool risks. you need a designed settlement mechanism before putting expiring assets into an ordinary pool.

even when only the refund right expires, the market loses that redemption option at the deadline. price can change sharply.

the interesting part is giving buyers a defined exit right, backed by assets and enforced through code.

more token launches should experiment with that.

expiry makes it even more interesting

Expiry Makes It Even More Interesting

Now combine refunds with expiration.

The refund right could expire after seven days while the token itself continues trading. Or the token itself could expire. Those are two completely different designs.

From there, you can imagine seasonal money. Tokens issued for one gaming season, one event, one community campaign, one grant cycle, one election, one loyalty period, or one speculative game.

At the end of that period, balances could expire, transform, or settle into something else.

Now you are no longer designing "another ERC-20." You are designing new versions of money.

This is the part of crypto I think has barely been explored.

what hooks actually are

i keep mentioning hooks, so let me explain them simply before the next chapters lean on them.

In simple words hooks are like Wordpress plugins.

Using hooks you can set conditions before after swap of a token in a liquidity pool.

When you set conditions.

You can set token rules.

You can create new token rules.

You can explore wild ideas.

in uniswap v4, a hook is a contract attached to a liquidity pool that can run custom logic at specific moments: before or after a swap, before or after liquidity is added or removed. that means a pool can charge dynamic fees, enforce time windows, route value to a vault, block certain behaviours, or record data, without forking the exchange. one important limit: a hook only controls what happens inside its pool. it cannot automatically impose rules on every transfer of a token everywhere. if you want rules that travel with the token itself, those belong in the token contract. good designs usually need both.

§ chapter 13

13

memberships, credits, and things you stop using

a lot of value in the world is locked inside things we bought and stopped using. a gym membership. a loyalty tier. a pile of AI credits. programmable tokens can turn that dead value into something portable.

the membership in your wallet

your next club membership could sit in your wallet. and when you stop using it, you could sell it.

onchain club membership.

hold a certain amount of membership tokens, and get access to exclusive opportunities.

sounds like what NFTs wanted to achieve.

but you can also use ERC-20 tokens for the same purpose, with a clear definition of how many tokens get you what.

hold 100 tokens.

unlock member-only prices.

hold 500 tokens.

get early access to product drops and private events.

hold 1,000 tokens.

unlock the highest membership tier, with priority bookings and exclusive experiences.

there is a token standard called ERC-8063 that formalizes this idea.

your token balance represents your membership level.

the business defines the benefits attached to each level.

buy or earn more tokens, and you can move up.

sell enough tokens, and you move down.

eventually, some traditional clubs might bring their membership and points programs onchain.

when they do, ERC-8063 is one standard they could use.

and you can explore the same thing for your offline or online business, especially D2C brands that want to reward retention and loyalty.

imagine a clothing brand.

customers earn membership tokens through purchases, verified referrals, or repeat orders.

as their balance grows, they unlock early access to collections, alterations, private shopping sessions, or better prices.

now imagine that brand partners with a sneaker store, a gym, and a café.

each business can recognize the same membership token and offer its own benefits.

your membership starts working across a network of businesses.

and if transfers are enabled, you could sell your tokens when you no longer want the benefits.

another customer buys them and joins at the corresponding tier.

but this creates an interesting distinction.

someone who buys 1,000 tokens can qualify for the same tier as someone who earned 1,000 through years of purchases.

so the token balance proves membership eligibility. it doesn't automatically prove loyalty.

brands that want to reward actual retention could add benefits based on purchase history or how long someone has held their membership.

there's another design choice too.

if spending your points reduces your membership balance, redeeming a reward could accidentally downgrade you.

a business could keep spendable reward points separate from the tokens used to determine membership.

ERC-8063 provides the membership framework. the business still has to build the rewards, redemption rules, and customer experience around it.

the opportunity is to make membership portable, transferable, and useful across more places.

but the reason people stay will still be the same.

the benefits have to be worth belonging for.

a secondary market for unused AI

your unused AI credits could be someone else's next viral video. so why can't you sell them?

AI credits are valuable to many.

but when you use different platforms, you end up leaving unused credits everywhere.

that's what happened to me.

i found unused credits sitting in my Seedance, HeyGen, and OpenArt accounts.

i paid for them.

i didn't use them.

money spent. utility left behind.

so then i thought: what if i could sell those credits on a marketplace at a discount?

i list my unused balance.

buyers place bids.

we discover a price that works for both of us.

someone gets cheaper AI access.

i recover some of my money.

would be cool, right?

and what if all of this happens ONCHAIN?

yes, it's possible to build.

there is a token standard called ERC-6909 that lets you create different token IDs within a single smart contract.

each ID could represent a different kind of service credit.

one for image generation.

another for video generation.

another for AI avatars.

credits with different providers, usage terms, or expiry dates would have separate IDs, so buyers know exactly what they're buying.

imagine a participating platform issues 1,000 image-generation credits to my wallet.

i use 400.

i list the remaining 600 for sale.

let's say those remaining credits would cost $60 when bought directly from the platform.

a buyer bids $40.

another bids $45.

i accept $45.

the marketplace contract transfers the credits to the buyer and the payment to me in the same transaction.

the buyer then redeems those credits to generate images, and the redeemed credits are burned.

ERC-6909 also allows specific spending approvals.

i could let the marketplace transfer only the 600 image credits i'm selling, without giving it access to my other credit types.

but there's one important catch.

the AI provider must support this system and honour the credits when a new owner redeems them.

i can't simply turn my existing Seedance, HeyGen, or OpenArt balance into tokens and assume those platforms will accept them.

the token standard handles balances, transfers, and permissions.

the marketplace needs bidding and payment logic.

the provider integration makes the credits usable.

and that's where the real business opportunity is.

build this with AI providers willing to make their credits transferable, or operate an AI service gateway that issues and honours its own credits.

over time, buyers could bid based on the provider, model quality, service demand, and time left before expiry.

credits about to expire might sell at a deeper discount.

credits for an in-demand service might attract better bids.

a secondary market for unused AI access.

i already paid to create something.

if i'm not going to create it, let someone else buy that chance.

> *i already paid to create something. if i'm not going to create it, let someone else buy that chance.*

both ideas share the same design lesson. the token standard is the easy part. the hard part is the real-world promise behind it: the café that honours the tier, the AI provider that honours the credit. crypto can make a promise transferable. it cannot make it true. that line will keep coming back.

14

one nft, many lives

nfts had a bad few years. most of them deserved it. but the technology underneath is far more interesting than the profile pictures that made it famous.

imagine buying one NFT that you can wear as your identity, play as a character, read as a comic, and listen to as a song.

think about this NFT collection called $MANY.

very simple name, but what it does feels magical.

on a marketplace, you trade it like a typical NFT.

inside a compatible online game, the same NFT supplies a 3D model you can use as your character.

open it in a compatible reader, and it presents a comic.

open it in a music app that supports it, and it presents the character's soundtrack.

one NFT.

multiple assets.

different experiences depending on where you use it.

you don't need to move it out of your wallet every time.

the supporting app reads the NFT and loads the asset relevant to that experience.

an SVG becomes your PFP.

a 3D model becomes your gaming avatar.

a PDF becomes your comic.

an audio file becomes your soundtrack.

a video becomes your animated origin story.

all connected to the same token.

there is an Ethereum standard called ERC-5773: Context-Dependent Multi-Asset Tokens.

it extends ERC-721 so a single NFT can have multiple associated assets, with preferences controlled by its owner.

now imagine building an original character around this.

you buy MANY #108.

that character has a face, a body, a story, a voice, and a world it belongs to.

an illustrator creates its portrait.

a writer develops its comic.

a musician creates its theme song.

a game developer builds a playable version.

those contributions become different ways to experience the same character.

and the character can keep evolving.

the issuer could propose a new animation, an updated model, or the next chapter of the comic.

the owner, or an approved asset manager, accepts or rejects the proposed addition or replacement.

you could also prioritize which assets supporting apps should display.

so the NFT you buy today could gain new creative material over time, without requiring a separate NFT for every format.

there are two things to get right.

first, game compatibility still requires developers to support the assets. attaching a 3D model doesn't make it playable in every game.

second, attaching a book or song doesn't automatically give the holder its copyright or commercial rights.

if $MANY allows holders to make merchandise, publish adaptations, or license the character through an IP marketplace, those permissions need to be clearly defined separately.

the files must also be created and hosted. the standard connects them; it doesn't generate them.

that's where the business gets interesting.

you could build a character studio where each release comes with a complete creative bundle, then expand it through games, music, comics, and licensed collaborations.

one character could become something you collect, something you play, and something you build stories around.

$MANY.

one NFT. many ways to bring it to life.

($MANY is a fake token used just for the name to explain this erc)

provenance for the age of AI

and the same thinking applies to the flood of AI-generated content.

you saw an AI-generated scene you couldn't stop watching.

what if you could buy the work, along with a verifiable record of how it was created?

the internet is filling up with AI-generated content.

and there is a reason for it.

AI unlocks dreamlike possibilities when you create characters and build scenes around them.

> a city floating inside a sleeping giant.

> a character walking through different versions of their childhood.

> an entire universe built around an emotion.

> things you could imagine but couldn't easily produce before.

> now some of these generations are difficult to replicate.

sometimes you need the exact prompt.

sometimes even the exact prompt won't give you the same output.

because the result can also depend on the model version, seed, settings, reference inputs, and the workflow behind it.

the prompt is only part of the creative process.

the artist still experiments, selects, rejects, edits, and connects the outputs into something people want to watch.

and when a generation attracts attention or becomes useful to someone else, there could be a market for it.

> a game developer might want the character.

> a musician might want the visual.

> a brand might want to license the scene.

> a collector might want a particular edition.

so artists should have ways to sell their work, license its use, or share the generation recipe under clearly defined terms.

this is where ERC-7007 gets interesting.

it is called the Verifiable AI-Generated Content Token standard.

it extends NFTs with a way to connect an AI-generated output to its input and verify that relationship against a specified model.

the NFT can reference the content and its generation metadata.

the verification system checks the computation behind it.

there are two approaches.

zkML uses a cryptographic proof to verify that the supported computation produced the output.

opML uses an optimistic process where a submitted result can be challenged before it becomes final.

both require actual verification infrastructure around the model. uploading an image and adding “AI-generated” to its metadata isn’t enough.

imagine an artist releasing a collection through a defined model.

each piece comes with its generation input, a reference to the output, and verification that connects them.

a buyer can inspect the provenance instead of relying only on the seller's claim.

the artist could separately offer a commercial licence, a workflow package, or access to the model where they have the right to provide it.

those are different products.

buying the NFT doesn't automatically give someone copyright, model weights, or permission to use the character in a movie.

the terms need to say what the buyer actually receives.

and verification has a boundary too.

if the artist edits the output afterward, a proof of the original generation doesn't automatically verify every edit in the finished scene.

but that still leaves a useful foundation.

a marketplace where artists can sell distinctive AI works with clearer provenance, and buyers can understand what they are collecting or licensing.

the file can still be copied.

the collectible edition, documented generation history, and licence give people specific reasons to buy.

ERC-7007 doesn't make every generation valuable.

it gives creators a way to support the provenance of the ones people already care about.

what you actually own

notice the question that keeps returning in both of these ideas: what exactly does the buyer receive? the file, the edition, the licence, the copyright, the model, the right to use a character in a film. these are different things, and the token only carries what the terms say it carries. most nft disasters were not technology failures. they were promise failures, where people thought they bought rights they never got. if this market comes back, it should come back with the rights written clearly on the box.

15

the casino can be redesigned

we have walked through serious standards. now let me talk about the part of crypto everyone pretends to be above but secretly lives inside: memes and speculation.

People love saying crypto is a casino. Fine. Maybe it is.

But casinos also have rules, and rules can be changed.

Right now, one rule remains incredibly powerful: people who bring more people into the casino make the most money. Attention sits above almost everything.

But the casino doesn't have to remain maximally extractive.

You could experiment with refund mechanisms, better distribution models, progressive sell taxes, anti-sniper rules, privacy, purpose-bound money, expiring assets, reputation-based access, conditional liquidity, commitment rewards, and different forms of coordination.

The goal is not to remove speculation. Speculation is probably one of crypto's strongest distribution mechanisms.

The goal should be to make speculation less stupid.

If people are going to gamble anyway, at least give them better games. Games with understandable rules. Games with different ways to win. Games where newcomers are not automatically exit liquidity.

That alone would be a meaningful improvement.

token breeding

next gen memecoins.

you select a theme lets say protest coins.

and than you launch a coin representing a protest slogan.

and you issues 10M coins.

you as a dev can send these coins to qualified wallets.

thats your simply erc20 you all know that, but what i am gonna propose here will blow your mind.

you can actually allow holders to burn tokens, and mint new tokens everyday.

you can decided that rule.

where if the genesis token is $PEACE and they have an option to burn or sell that token and buy or mint new token which would continue the lore and theme of coins.

in that context, we could actually build launchpad for different kind of theme based memes.

where new coins are getting born from the old one, its more like token breeding.

where you can burn and mint.

you as a holder gets the right which generally goes with dev.

here dev can simply write custom hooks, and we can allow more people to become part of this experiment.

In ETH, ERC7837 is that standard.

In Solana, Using Solana token extension you can achieve the same outcomes.

and now you've hooks.

so what you're waiting for.

its time to launch this experiment or find out who is doing that as part of your alpha hunting routine.

Cheers.

new slot machines

Best part here is atleast 100+ such new wild token rules have been introduced by various ERC authors in the past.

The market simply needs new narratives.

Micro narratives to keep the extraction window open for longer time.

You are in casino.

Casino owners need new ways of slot machine in simple words.

Now anyone can become a dev and build those slot machine, new kind of slot machine.

Why because AI will code for you all you need to give ideas.

Hooks make them easy.

Ideas are there in ERC docs.

You just need to read make judgment and build new slot machines for casino.

Some of the slot machine can bring fair outcomes to gamblers.

You are gamblers don't offend.

I am gambler too when I am in casino.

How to gamble in a way where we don't get front run by insiders is the whole question.

These experiments are answer to it.

You can actually make 10x on your folio if you begin to research new token designs, and figure out which design could grab more memetic value, and you just need to be early to those projects which integrate those designs through hooks. This is not rocket science. It takes less than 30 mins to figure out a sheet with 40-50 memetic token designs, and simply track everyday which is aligned with those tokens. Crypto is 30 mins game. Not more than that. Those who realize will make the money while touching grass everyday those who can't will keep complaining.

i stand by the excitement in that post, but let me add the warning i would give a friend. a new token design is not automatically a better game. some designs make speculation fairer. others just make extraction more creative. the test i use is simple: does this rule give newcomers a better chance, or does it give insiders a new hiding place? if you cannot answer that, you are not early, you are the liquidity.

we need to expand what money means

Imagine a world where a meme token becomes money inside its own ecosystem.

You can trade it, spend it, earn it, use it privately, use it to access communities, use it to coordinate agents, use it for services, use it in prediction systems, use it for reputation, and use it to make commitments.

At that point, the meme stops being just an asset. It becomes economic infrastructure.

This is the direction I think crypto needs to explore.

Not one universal money, but thousands of programmable monies with different rules, communities, philosophies, risk models, privacy models, and forms of commitment.

Crowds would choose which rules they want to participate under.

That is a much more interesting future than every token behaving exactly the same way.

the next developer might just be someone with ideas

AI changes one more important thing: building these experiments is becoming easier.

You don't need to spend five years becoming the best Solidity engineer before thinking about token design. You can study standards, understand the rules and risks, write the logic, use AI to turn the idea into an implementation, test it, break it, improve it, and launch experiments.

The bottleneck increasingly moves from **can you code?** to **can you think of interesting rules?**

That could create a completely different generation of crypto developers. Developers who start with philosophy, economics, game design, psychology, and coordination, then use AI to translate those ideas into code.

And I think that is where crypto becomes interesting again.

Stop asking only what token will pump.

Start asking what new rule a token can introduce.

Because once money becomes programmable, token design becomes social design. Whoever designs the rules designs the game.

The next big crypto primitive might not be another chain, another exchange, another lending protocol, or another meme launcher.

It might simply be **a better rule for money.**

> *whoever designs the rules designs the game.*

§ part 5

part five

the social trilemma

identity, privacy, reputation

for years the most famous problem in crypto was the blockchain trilemma: you can't easily maximise decentralization, security, and scalability all at once. we never fully solved it. we just stopped talking about it.

We failed to solve the system level blockchain trilemma.. and now we don’t even talk about it.. most of new people don’t even know this word.. and we are all fine with it.. but for how long? .. you got a new trilemma to solve now.. social trilemma.. around identity, reputation and privacy.. and most of folks might have no idea about your public key is your identity you just need to translate it into more human readable form and than you get shareable identity but identity without persona and verifiable data is more like dead body.. so think more. Shill less the space needs more thinking.. we got all kind of wrong thought leaders at the top.. all wrong messaging.. no values left… and we can’t sit silent.. we will speak.. who is is we? Me and the one who cares from the community..

this part is the heart of the book. it argues that the real bottleneck to crypto adoption is no longer financial infrastructure. it is our inability to build a society around it. and a society needs three things money alone cannot give it: a way to know who someone is, a way to protect what they don't want to share, and a way to trust what they have done.

§ chapter 16

16

a society, not just an economy

we spent more than a decade building a financial system on blockchains. but what if the biggest bottleneck to crypto adoption is not financial infrastructure anymore? what if it is our inability to build a society around it?

i have been thinking about this question for quite some time. whenever we talk about the bottleneck problems of ai, people discuss computation, energy, data, and intelligence. but when we talk about the bottleneck problems of crypto, we usually end up discussing scalability, liquidity, regulation, and user experience.

these are important problems, but i think we are missing something much more fundamental. we are trying to build a new economic system without solving the problems that allow humans to organize themselves into a society. and those problems are identity, privacy, and reputation.

i call this the social trilemma of crypto. it is not the classical blockchain trilemma of decentralization, security, and scalability. it is a different problem, and i believe solving it could unlock an

entirely new generation of crypto applications.

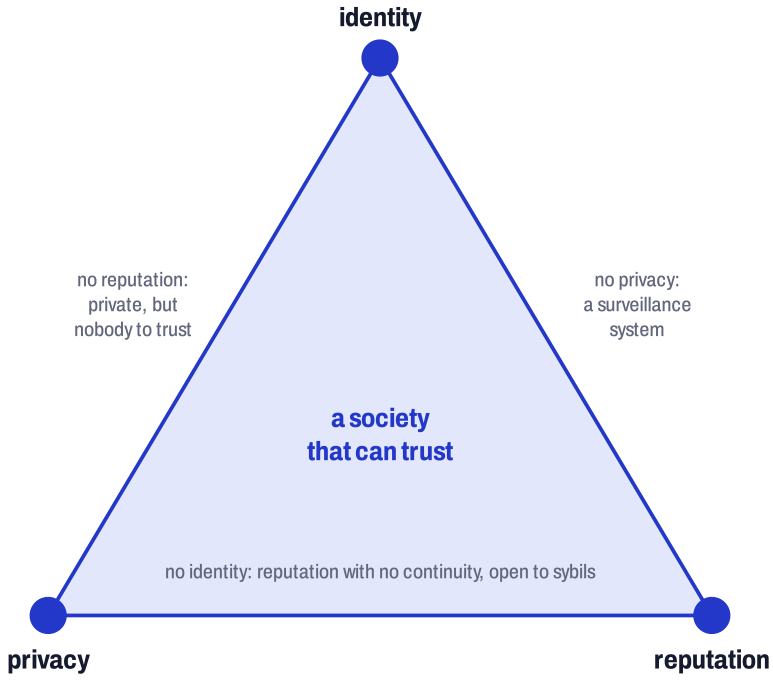


fig. 1 the social trilemma: three corners, one goal

remove any one corner and the other two break in a predictable way. that is the whole argument of this piece in one picture.

cryptography already gave us an identity

let's start with something very simple. when you create an ethereum wallet, you generate a private key and derive a public key and address from it. that address becomes an identifier through which you interact with the blockchain: you can receive money, send

money, own assets, participate in governance, and interact with thousands of applications. in a way, cryptography has already given you the foundation of a digital identity.

but there is a problem: a blockchain address is not human-readable. this is where ethereum name service (ens) comes into the picture. ens allows us to translate complicated blockchain addresses into names that humans can understand, so instead of sharing a long hexadecimal address, i can share something like hitesh.eth.

now imagine a society where millions of people use these names as their primary digital identifiers. they can exchange money, communicate, build communities, and interact with applications using the same identity. we have already built the foundation of an internet-native society. but we have not solved the identity problem, because knowing what someone calls themselves is not the same as knowing who they are.

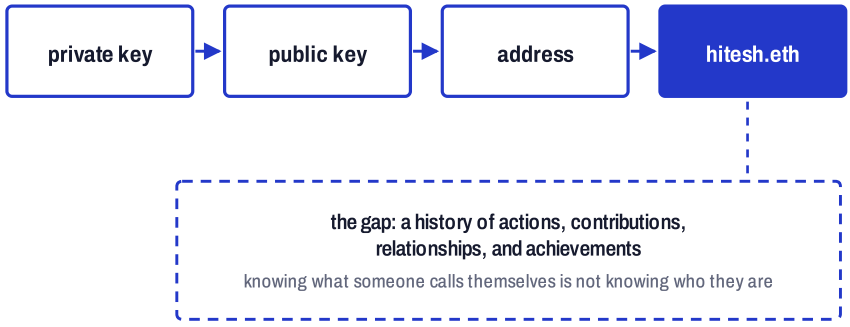


fig. 2 from private key to ens name, and the gap after it

a person can create an ens name and claim to be anyone. i can claim that i have been contributing to ethereum since 2016, and someone else can make the same claim. how does society distinguish between

us? how do we establish that a particular identity represents a real history of actions, contributions, relationships, and achievements? that is where the actual identity problem begins.

identity and persona are two different things

i think we need to separate identity from persona. identity is the identifier through which you exist in a digital society. persona is the collection of information, behavior, contributions, and relationships that tells society something about you.

think about the physical world. you have a name, a passport, and a driving license. you might have academic qualifications, professional experience, and relationships with different institutions. these are different containers of information about the same person, and they serve different purposes: your driving license proves something different from your university degree, your degree proves something different from your professional reputation, and your professional reputation might tell people something that neither document can establish.

now imagine bringing this concept into crypto. your blockchain identity should be able to aggregate different kinds of verifiable information. take someone who has been participating in crypto for ten years. they have written research articles, contributed to open-source projects, participated in governance, funded startups, mentored developers, and attended community events. some of these activities happened onchain, others happened offline, but both contributed to the development of the ecosystem.

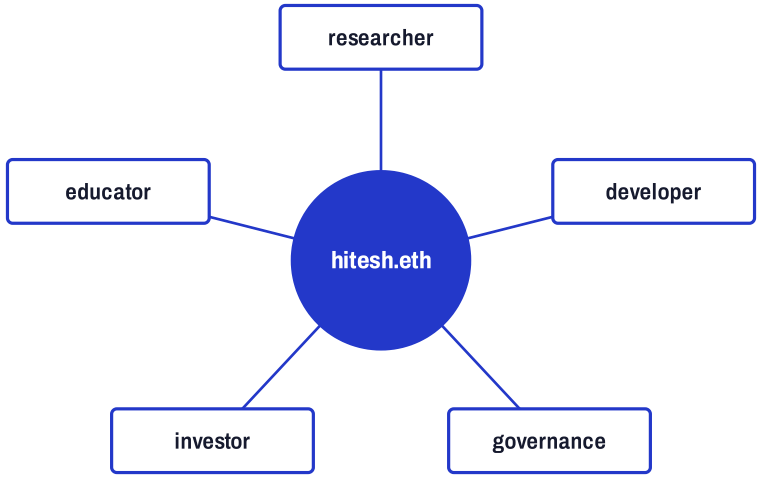


fig. 3 identity vs persona: one identifier, many personas

why should the blockchain only recognize the transactions made by this person and ignore the intellectual and social contributions they have made? why should someone who traded $100 million automatically appear more important than someone who spent ten years educating thousands of people about cryptography? this is where i believe the next generation of nfts and identity infrastructure becomes interesting.

nfts as containers of human history

we have spent years discussing nfts as profile pictures, collectibles, and speculative assets, but their potential goes much further. an nft can represent a persona. it could include a visual representation of you, perhaps even a 3d model. it could reference your contributions,

credentials, memberships, achievements, and relationships. it could become an interface through which society understands the person behind a blockchain identity.

and we don't necessarily need to invent every technical standard from scratch. ethereum already has relevant building blocks: erc-721 for unique tokens and metadata, erc-5192 for locked, nontransferable nfts, erc-5484 for soulbound tokens, ethereum attestation service for verifiable claims, and erc-6551 for token-bound accounts. none of these standards individually solves the complete identity problem. but together with decentralized identifiers, verifiable credentials, encryption, and selective disclosure, they provide a starting point.

imagine a nontransferable identity credential linked to your ens name. behind it is an evolving collection of verifiable claims: your research contributions, development history, community participation, offline contributions, organizational relationships, and public achievements, and potentially your private credentials, which you can reveal only when necessary.

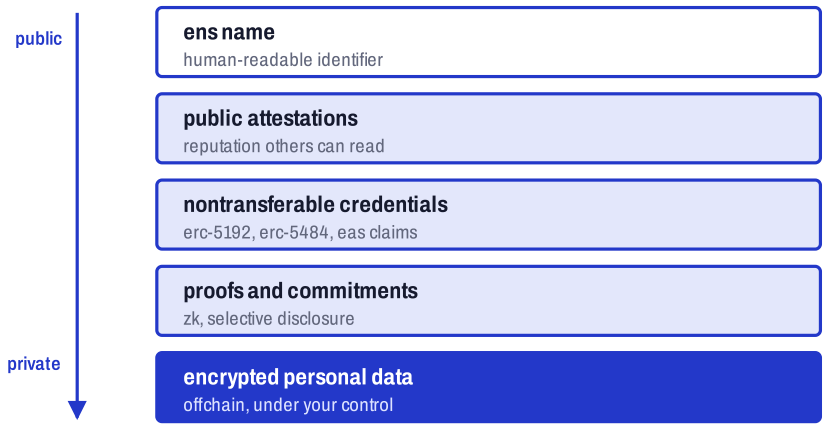


fig. 4 the identity stack: public on top, private at the base

importantly, the actual personal information does not need to live publicly on a blockchain. sensitive information can remain encrypted offchain, while cryptographic proofs, commitments, or references help verify selected claims. the identity token becomes a pointer to a much richer system: one identifier, multiple personas, thousands of contributions. and that brings us to the second bottleneck.

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trust without transparency

suppose we succeed in building this rich identity system. every contribution you have made is associated with your identity: every financial transaction, every organization you interacted with, every community you joined, every credential you earned. sounds powerful, right? but now imagine that everyone can see everything. your financial history becomes public, your relationships become public, your personal behavior becomes public, and anyone can analyze your entire life.

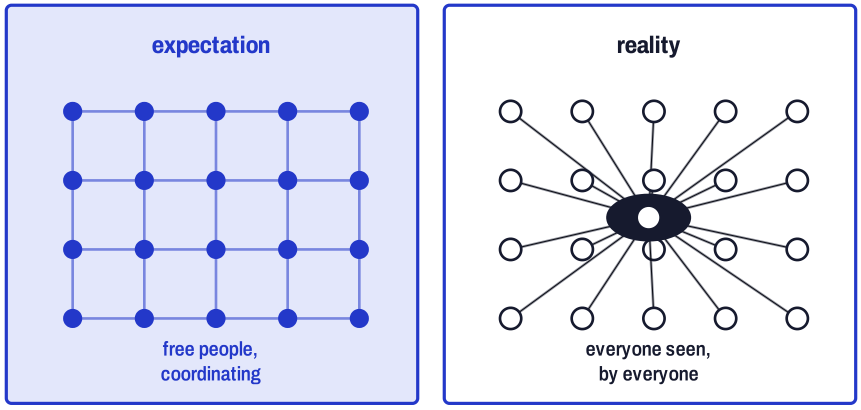


fig. 5 expectation vs reality: identity without privacy

we would have created the most powerful surveillance infrastructure in human history while claiming to build a system for human freedom. that would defeat one of the original purposes of crypto. this is why privacy cannot be treated as an optional feature.

privacy should be a fundamental right of the individual, while transparency should be an obligation attached to specific public responsibilities. these two ideas are not contradictory. when you make a private payment to another person, society does not automatically need to know the amount, recipient, or reason. but when you receive money from a public treasury to perform a public responsibility, society has a legitimate interest in verifying that the money was used according to the agreed rules.

the distinction is important. we need systems that can make claims verifiable without making every underlying piece of information visible. imagine proving that you have contributed to ethereum for five years without revealing every wallet you have ever controlled, proving that you meet the requirements to participate in a

community vote without revealing your real-world identity, or proving that you fulfilled a contractual obligation without publicly exposing confidential business information.

this is where zero-knowledge proofs, selective-disclosure credentials, and privacy-preserving identity infrastructure become useful. but cryptography alone cannot decide what information should be public and what should remain private. those decisions involve social rules, consent, and governance, which brings us to the third problem.

reputation is the missing social infrastructure

we have identity. we have a way to build personas. we have a way to selectively share information. now the question becomes: how does society decide whom to trust? in the physical world, reputation develops through repeated interactions. you know someone because you have worked with them, someone else trusts them because they delivered on a commitment, and another person recognizes them because of their contribution to a particular field. reputation is not just a number. it is a social consensus formed around historical behavior, and this is exactly what we need to build in crypto.

but there is a fundamental difference between a reputation system and a popularity contest. someone with one million followers is not necessarily more knowledgeable than someone with one thousand followers. someone who owns $10 million in tokens is not necessarily more trustworthy than someone who owns $10,000. someone who has participated in thousands of transactions is not necessarily a meaningful contributor to the ecosystem. yet crypto has repeatedly confused wealth, activity, and attention with credibility.

we have built systems that measure how much capital someone controls, but very few systems that measure how much value someone has contributed to society, and i think this is one of the biggest mistakes we have made. reputation should emerge from verifiable contributions, peer validation, and historical behavior.

it should also be contextual. a person might have an excellent reputation as a developer but little experience managing public funds. someone might be a brilliant researcher but a poor business operator. someone might have contributed to ethereum for ten years but know very little about zero-knowledge cryptography. we should not collapse all these different dimensions into one universal reputation score. instead, we should build a system where different reputations coexist under the same identity.

proof of knowledge

this is where i want to introduce another idea. when we discuss validators in crypto, we usually think about people or institutions staking capital to secure a network. but what if we built another kind of validator network, one where people become validators because of their knowledge, contribution history, and reputation? i call this concept proof of knowledge. it is not a replacement for proof of stake or the consensus mechanism securing ethereum. rather, it is a social verification layer operating on top of decentralized infrastructure.

imagine a researcher claiming that they have published 100 original articles about ethereum. the claim could be submitted to a verification network, and validators who have demonstrated expertise in research, cryptography, or the ethereum ecosystem could review the evidence. they could examine publication history,

timestamps, originality, attribution, and external references. their conclusions would become attestations attached to the researcher's identity.

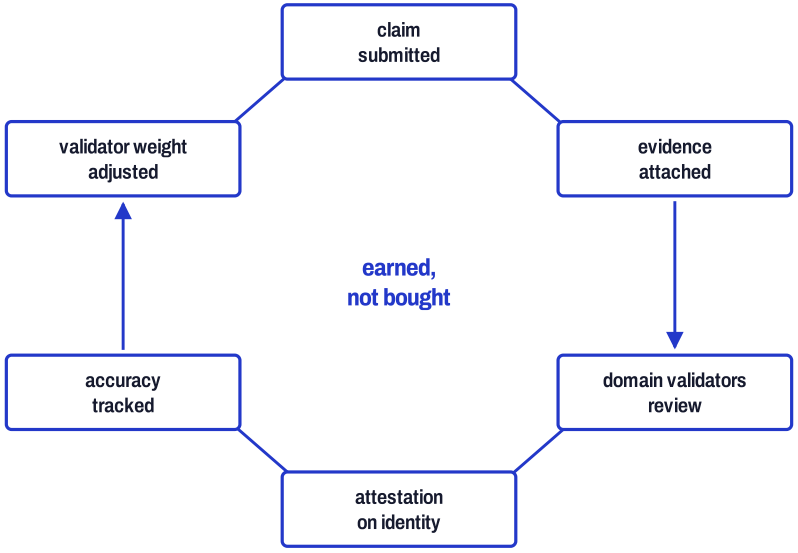


fig. 6 proof of knowledge: claim, review, attestation, and the accuracy loop

but not every validator would have equal influence. someone with a ten-year history of authentic contributions in that domain might carry greater weight than someone who joined yesterday. someone whose previous attestations were consistently accurate might gain more credibility, and someone repeatedly caught validating false claims could lose their authority. importantly, validators would not need to become wealthy to participate. their primary qualification would be demonstrated knowledge.

however, longevity alone should never guarantee authority. otherwise, we simply replace financial aristocracy with an intellectual aristocracy. new contributors need a path to earn trust, established validators must remain challengeable, and no single group should be able to permanently control the system. this is how we could begin building a reputation economy where credibility is earned through contribution rather than purchased through capital.

who decides what is public and private?

this is where the idea becomes more philosophical. suppose we create a society where identities are verifiable, contributions are recorded, and reputation is portable. who creates the rules? who decides which information must be publicly disclosed? who decides whether someone has violated an agreement, and who decides whether a validator has acted dishonestly?

my answer is that we should experiment with voluntary governance systems. people should be able to organize into communities with clearly defined rules. they should understand those rules before joining, and they should have the ability to challenge decisions and leave those communities without losing control of their fundamental identity.

imagine a community of researchers. they collectively agree that anyone claiming authorship of a research paper must provide evidence. they establish standards for originality, citation, and peer review, select validators based on proven expertise, and define a transparent appeals process. now imagine another community of developers who use entirely different standards to evaluate contributions, and a third community that evaluates public-goods funding. the underlying identity remains portable, but the reputation rules differ by context.

this creates a network of voluntary institutions rather than one central authority deciding who deserves trust. the smart contracts could enforce objectively verifiable rules, while human review handles disputes involving interpretation, conflicting evidence, or fairness. it is important not to confuse this with replacing national law: voluntary community rules still need to respect legal rights, privacy protections, and due process.

how could this actually work?

let's imagine a practical architecture with five layers, each resting on the one below it.



fig. 7 the architecture: five layers, identity at the base

the first is a portable identity layer. users control wallet-based identifiers, connect ens names, and manage key rotation or recovery, because identity should not disappear permanently just because someone loses a private key. the second is a credential and persona layer, where people collect verifiable claims about their

work, achievements, qualifications, and participation. those credentials can be issued by institutions, communities, other individuals, or automated systems.

the third is a privacy layer. personal information remains under user control where technically possible, and selective disclosure and zero-knowledge proofs allow people to demonstrate specific facts without exposing unrelated data. the fourth is a reputation layer, where domain-specific reputation develops through attestations, evidence, historical accuracy, and peer review. reputation is portable, but it is not blindly accepted everywhere.

the fifth is a voluntary governance layer. communities define verification standards, appoint or elect reviewers, establish challenge mechanisms, and use smart contracts for rules that can be enforced reliably. the goal is not to create a single global identity authority. the goal is to create interoperable systems of trust.

the hardest part is preventing capture

every reputation system creates incentives, and wherever incentives exist, people will try to manipulate them. someone could create thousands of identities to validate their own claims. a group of validators could coordinate to reward each other. established contributors could prevent newcomers from gaining influence. organizations could issue meaningless credentials, people could buy fake attestations, and governments or corporations could use reputation infrastructure to discriminate against individuals.

these are not minor edge cases. they are fundamental design problems. any serious implementation needs to address sybil resistance, validator collusion, evidence quality, revocation, appeals, data minimization, and governance capture.

there is also an important tension between permanent records and human forgiveness. should a mistake someone made ten years ago define their reputation forever? should a person have the ability to recover from past behavior? can a system preserve the integrity of historical evidence while allowing rehabilitation?

a reputation system that never forgets anything could become a digital caste system, and that would be the opposite of the open society we are trying to build. i think we need to distinguish between preserving verifiable historical records and permanently punishing people for those records. reputation should be capable of evolving.

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infrastructure for a society

now imagine that we solve these problems reasonably well. what becomes possible? a researcher could obtain funding based on a verifiable history of intellectual contributions. a developer could receive opportunities based on demonstrated expertise rather than follower count. a community could select reviewers based on knowledge instead of token holdings. a small business could build portable credibility without depending entirely on a centralized platform, and people could enter into agreements with strangers because relevant credentials and previous commitments are verifiable.

and ai agents could eventually operate with delegated identities, permissions, and reputation histories of their own. an ai agent managing money would not necessarily need to expose every transaction publicly, but it could be required to prove that it operated within a spending limit, followed a mandate, and did not violate a particular contractual rule. its owner could determine what information is disclosed to counterparties, auditors, or the public,

subject to agreed obligations and applicable law. the agent's reputation could be linked to its own behavior and the accountability of its operator.

this is where identity, privacy, and reputation begin to converge. and this is where crypto starts looking like infrastructure for a society rather than just infrastructure for financial speculation.

capital has an identity. contributions often don't.

this is the part i find most interesting. in crypto, if someone holds $100 million worth of tokens, we can often observe that wealth onchain. if someone moves millions through defi protocols, we can analyze their activity. if someone owns a valuable nft collection, we can verify ownership.

but what about someone who spent ten years teaching people about decentralization? what about the developer who maintained an important open-source library without ever launching a token, or the researcher who helped thousands of people understand the risks of a protocol, or the early community members who helped an ecosystem survive when almost nobody believed in it? where is their contribution recorded? where is their reputation? why have we built so much infrastructure to recognize capital but comparatively little infrastructure to recognize human contribution?

i think this is one of the reasons crypto culture has gradually shifted toward extraction. the system is very good at recognizing what you own, but much less developed at recognizing what you have given. when ownership becomes the dominant signal of importance, people naturally begin optimizing for ownership. when attention becomes the dominant signal of reputation, people begin optimizing for attention.

but if we build systems that recognize authentic contribution, perhaps we can begin changing those incentives. not by asking people to become morally superior, but by designing an economy where contributing to society creates durable, verifiable value.

the trilemma, again

so let's return to the three bottlenecks. identity allows people to establish continuity and make verifiable claims about themselves. privacy gives individuals control over what they reveal and protects them from unnecessary surveillance. reputation allows society to evaluate contributions, behavior, and credibility over time. all three are necessary, and each combination that leaves one out fails in its own way.

this is not a mathematical impossibility theorem. it is a design challenge involving competing social and technical requirements. we should stop treating these problems as separate industries, because they are deeply interconnected.

the next applications might not begin with a token

for years, we have approached almost every problem in crypto with the same question. how do we tokenize this? how do we create liquidity around it? how do we incentivize users to participate? how do we launch a market?

but perhaps we need to ask a different set of questions. how do we help strangers establish trust? how do we allow individuals to prove their contributions? how do we preserve privacy while enabling accountability? how do we build voluntary institutions that can enforce meaningful commitments? how do we make reputation portable across applications and communities? and how do we

ensure that the people who contributed to building this ecosystem are not forgotten simply because they no longer control significant capital or attention?

these questions might not create the next speculative frenzy, but they could create the foundation for applications that people actually need. most people do not wake up every morning thinking about blockchain scalability. they think about earning a living, finding trustworthy people, protecting their privacy, building relationships, receiving recognition, and participating in communities. crypto becomes relevant to ordinary people when it helps them solve these problems, not when we simply convince them to buy another token.

building society

we started with bitcoin because we wanted money that could operate without requiring permission from centralized institutions. then we built ethereum because we wanted programmable agreements and decentralized applications. we built defi because we wanted financial markets that could operate through open protocols, nfts because we wanted digitally verifiable ownership, and daos because we wanted new ways of coordinating people. each innovation expanded what blockchains could do.

but we have not yet connected these pieces into a sufficiently mature social infrastructure.

perhaps the next important breakthrough will come from bringing these pieces together: an internet where identity is portable, where personal data is protected, where contributions can be verified, where reputation is earned and challengeable, where rules emerge

from voluntary cooperation, and where smart contracts help enforce commitments without requiring everyone to trust the same central authority.

the original promise of crypto was not merely to create new assets. it was to create new possibilities for human coordination. and coordination at scale requires something more than money. it requires trust: not blind trust in an institution, not trust based entirely on wealth, not trust based on popularity, but trust that can be established through evidence, history, consent, and verifiable behavior.

maybe we have spent the last decade building the financial infrastructure for a decentralized economy, and maybe the next decade should be about building the social infrastructure for a decentralized society. because an economy can exist without a shared society, but a society cannot function for very long without ways to establish identity, protect privacy, and recognize reputation. until we solve these three problems together, i believe crypto will continue to struggle to become something much bigger than the markets we have created around it.

we have learned how to make money programmable. now we need to learn how to make trust verifiable without making human beings transparent. that might be the real bottleneck to crypto adoption.

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contribution without a record

the trilemma essay ended with a claim: we built an economy where capital has an identity but contribution doesn't. this chapter shows how that played out in the real history of DAOs and airdrops, the two experiments that were supposed to reward contribution.

It all looked fine in theory, but when we started experimenting with these ideas, we discovered much deeper execution-level problems.

One of the biggest problems was how we reward contribution.

We wanted to create an economy where contribution mattered more than capital, but when we introduced DAOs and governance tokens, capital started winning over contribution. People who held more tokens gained more influence than those who spent years building, educating, and growing those ecosystems.

We decentralized ownership without necessarily decentralizing power.

And when people realized that capital had more influence than contribution, they naturally started taking the side of capital.

Then came airdrops, where we tried to reward participation and contribution. It was a beautiful idea because ordinary users could finally become stakeholders without investing significant capital.

But it created another problem.

People started participating not because they believed in the network, but because they expected rewards. Contribution became something people started manufacturing through artificial transactions, multiple wallets, and meaningless activities.

We started rewarding what was measurable rather than what was meaningful.

Someone generating a thousand transactions could receive more rewards than someone who spent a year educating hundreds of people about the same ecosystem.

And when people received rewards, many didn't realize the value of what they had received. They wanted more and more.

Those who received less felt cheated, and communities built around shared beliefs gradually became communities driven by financial expectations.

We created an economy that rewarded participation but failed to create a culture that valued contribution.

And this is where the whole space started becoming toxic.

Today, we have an overcrowded industry where many participants neither have enough capital to influence the networks nor the skills or incentives to contribute something substantial.

Meanwhile, a smaller group of people who have mastered the art of extraction through capital, influence, and distribution continue accumulating more wealth.

Capital gives them influence, influence gives them access, access creates better opportunities, and those opportunities help them accumulate even more capital.

The irony is that we wanted to build an economy where contribution could challenge the dominance of capital, but we ended up creating another economy where capital continues to dominate contribution.

And I think the problem is much deeper than tokenomics.

We built systems to verify ownership of capital but never built equally strong systems to verify the authenticity, history, and reputation of contribution.

We can verify that someone holds a million dollars in a wallet, but we struggle to verify that someone spent five years making an ecosystem better.

This is why I believe the next evolution of crypto needs to focus on identity, reputation, and contribution.

We need systems where people can build verifiable histories of their contributions, where reputation cannot simply be purchased with capital, and where governance recognizes expertise, contribution, and community trust alongside financial ownership.

Of course, reputation shouldn't become another form of aristocracy where early contributors permanently control everything. New contributors should have equal opportunities to build influence through meaningful work.

Because the original promise of crypto wasn't simply to distribute tokens. It was to distribute power and economic opportunities.

goodhart's law, onchain

there's an old idea in economics called goodhart's law: when a measure becomes a target, it stops being a good measure. airdrops are the purest onchain example i know. the moment people learned that transaction counts, bridge volume, and wallet age were being measured, those numbers stopped measuring belief and started measuring farming. we rewarded what was easy to count, so people produced what was easy to count.

the fix is not to measure nothing. it is to measure things that are expensive to fake: peer-validated work, demonstrated skill, kept commitments, outcomes verified over time. that is exactly where identity and reputation come back in, and why i keep saying they are the bottleneck, not the nice-to-have.

§ chapter 20

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the privacy stack

we said privacy has to be a fundamental right, not an optional feature. this chapter is about what that means in practice, layer by layer, because "privacy" in crypto is usually sold as one product when it is actually a whole stack.

Another problem nobody wants to admit is that user education in crypto is terrible.

We saw what happened with the 2021 generation. People arrived, burned capital, found scapegoats, followed the wrong influencers, and learned the wrong lessons. Then many of those same people became influencers themselves, and the cycle continued.

This becomes dangerous because when enough people collectively lose money through scams, hacks, bad actors, and broken systems, regulators eventually get political permission to monitor everything. Every transaction, every wallet, every movement of money.

And once that happens, people suddenly start talking about privacy.

But privacy cannot simply mean hiding one stablecoin transaction. Privacy needs to exist across the entire stack: your IP layer, RPC, wallet, swaps, transactions, balances, identity, and interactions.

Selective privacy is not enough if three other layers are leaking everything.

So instead of asking, "What is the best privacy coin?", people should probably start thinking about the **privacy stack**. Which protocols solve which layer? Which pieces are missing? Where does metadata leak? Where are centralized operators still sitting in the middle?

Once you see privacy this way, the entire research process changes. Education around that stack could become one of the most important parts of crypto adoption.

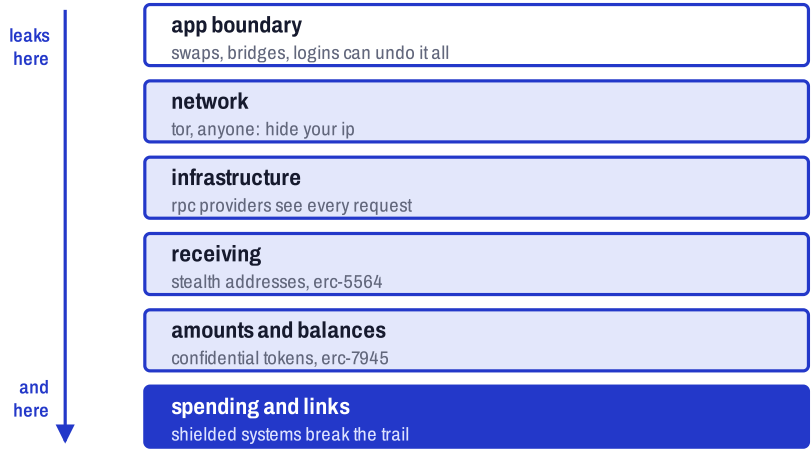


fig. 8 the privacy stack: every layer can leak

enough of all those privacy project shills, no one is actually providing end to end privacy.

we need an integrated privacy stack and its possible on ETHEREUM..

here's how we can achieve that.

the kind of wallet UI/UX you see these days feels magical to many of us who came from the 2017–21 era. we used to operate through some very shitty interfaces. you could have tokens worth thousands sitting in your wallet and still struggle to move them because you didn't have a few dollars of the right gas token. losing your seed phrase, without another backup, could mean losing access to everything.

now you have wallets with social recovery, fees payable in supported tokens, and sponsored transactions where you don't need to arrange gas upfront. ERC-4337 helped standardize infrastructure for these experiences through account abstraction. it didn't invent every feature, but it gave developers a common way to build programmable accounts and paymasters. recovery restores control of an account; it doesn't magically recover a forgotten seed. and gas still gets paid underneath, even when the user never touches ETH.

that's what interests me about standards. something that starts as a technical document can eventually change what millions of people expect from a wallet.

and another standard i think deserves more attention is ERC-5564: stealth addresses. @VitalikButerin is one of its authors. it allows someone to generate a fresh receiving address for you, without asking you to create and send a new address every time. the address remains visible onchain, but the cryptographic link between that address and your public identity is hidden from outside observers. the sender still knows whom they paid.

Fluidkey already brings stealth addresses into a usable product. its receiving flow uses fresh addresses, including an ENS name that resolves to a new address when queried. that is the kind of experience i want to see more of: privacy handled in the background while the user does something familiar.

think about why this matters. someone paying you for one piece of work shouldn't automatically get a window into your savings, other clients, donations, and spending habits. receiving money shouldn't require publishing your financial life.

but stealth addresses alone don't hide everything. the sending address and the amount can still be visible. and your next action can undo the privacy you just gained. if you fund a stealth address with gas from your publicly known wallet, you create a visible connection. Vitalik has discussed this exact problem: receiving privately is only part of the job; spending those funds privately matters too.

this is where the conversation needs more depth than "just add ZK."

a properly designed shielded system can use ZK proofs to demonstrate that a payment is valid without exposing the underlying private information. but adding a proof to an otherwise public transfer doesn't automatically hide the sender and receiver. the system has to be designed around which information is public, which is secret, and which links an observer can reconstruct.

ERC-7945 addresses another part of this problem. it defines a common interface for tokens supporting confidential balances and transfer amounts. the actual cryptography sits behind that interface. importantly, its transfer events still include addresses, so amount confidentiality and identity privacy remain separate problems. it also doesn't make existing ERC-20 balances private just because a wallet adds support for the standard.

so you can start seeing the architecture: stealth addresses for recipient unlinkability, confidential token implementations for amounts and balances, and additional shielded mechanisms for hiding the relationship between where funds came from and where they went.

but even that leaves another layer.

your wallet has to communicate with infrastructure. it checks balances, requests transaction data, estimates fees, and submits transactions. if those connections expose your IP address, protecting what appears on a block explorer only solves part of the problem.

Tor addresses network privacy by routing traffic through multiple relays, helping separate your originating IP from the service you contact. @AnyoneFDN is also building in this area, with a network and SDK for application integration. these belong in the wallet privacy conversation, though using similar routing concepts doesn't establish identical anonymity guarantees.

and routing alone cannot fix an application that identifies you through a login, tracking scripts, or a reused wallet connection. the browser, the wallet, the service receiving your requests, and the onchain transaction all need to be considered together.

that's why i think the real product opportunity is an integrated privacy stack.

imagine a wallet where receiving generates a fresh address automatically, supported assets use confidential balances, spending avoids reconnecting everything to your public wallet, and network requests travel through a privacy-preserving connection. the user sees a balance and a send button. the complexity sits underneath.

the SDK behind that experience would need to handle more than sending. it would need private balance discovery, fee payment, recovery, and ways for users to share selected transaction information when they choose. otherwise you might protect a payment and then expose the same accounts while restoring access or loading the dashboard.

there's also a difficult integration question: what happens when those funds enter an ordinary public dapp? if a swap, bridge, or withdrawal exposes identifying information, privacy can end at that boundary. an SDK should make those boundaries understandable instead of promising that every interaction is private.

that is the work i want more builders to talk about. making these components function together, testing where information leaks, and packaging the result so every dapp team doesn't have to become a cryptography research team.

my thesis is that Ethereum can deliver substantially better privacy through wallets and applications while deeper protocol work continues. ERC-4337 already demonstrates how meaningful changes to the account experience can be built without requiring consensus-layer changes. that doesn't prove a complete privacy stack is finished, but it gives us a useful direction.

we spent years making wallets easier to use. now i want that same effort applied to making them safer to use without exposing your entire financial identity.

privacy needs to survive the whole journey: receiving, holding, spending, connecting, and recovering. that's the wallet experience worth building.

the old internet is leaking

and this isn't only about wallets. the whole internet is a privacy problem waiting for better infrastructure.

the old internet infra is dying.

more hacks. more leaks.

then you ask what will happen to the internet now?

and the irony here is you're from CT.

CT should make posts about how next-gen crypto-powered infra can promise better security, privacy, and identity for the internet.

security could be achieved when more nodes secure the network, with more nodes keeping records of data.

privacy could be achieved through ZK, TEE, MPC.

and when you create a wallet, you get a public key. that's your identity on the new internet in its naked form.

you translate that into .eth or .btc or anything else you want.

then you use that identity to attach metadata with opt-in privacy.

and you use your identity everywhere you want.

composable and portable identity and apps.

powered by a public blockchain which has more "staked value" than anyone else. the staked value + how distributed that staked value is, with the right geographical diversity, will define how much better the next internet could become.

and since we're not talking about it.

that simply means we've been distracted by someone who wants a more centralised internet.

who is that someone: authoritarian states, deep state actors like companies that are building AI. some of the same companies built social media, a new reality for society which even influences our material reality to a very large extent.

The world will have more IOT devices.. most of these devices capture your sensitive data.. and you sensitive data which should be private to you it's getting sold to many companies already, and the trade of human privacy would keep getting bigger.. and that's more concerning than price action of $zec or community fight they are going through.. we need to think about how we can bring some level of security, and privacy guarantee back on the Internet.. and blockchain is the only tech that promise a perfect balance of security, privacy, scalability, identity atleast for now.. two other missing links are reputation and decentralization .. centralisation is dangerous for distributed networks .. so we need to do something where people get the problem than we should sell solutions while giving coownership of the onchain enterprise a builder decide to run and grow..

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a billion verified humans

india is the most interesting place in the world to think about identity, because we have already done something no blockchain has: verified more than a billion people. the question is what we let that verification become.

when you simply understand the fact that every KYC database is waiting to be breached.. that means your identity is pretty much exposed and there is no way to fix that damage.. so we need to build a new structure around identity, and that is only possible in right way when you create a public key on blockchain, translate that public key into human readable name like ens names and you add record to your end name while having control over what part of data you want to share with public, and what goes private.. we need to accelerate identity

india has a massive advantage when it comes to verifying humans at scale.

in march 2026 the government reported approximately 134 crore live Aadhaar holders and more than 17,000 crore authentication transactions.

we already have identity infrastructure operating at a scale most countries will never reach. but being able to verify yourself and having control over how your identity gets used are two different things.

your fingerprints are yours.

your eyes are yours.

but the database holding their digital representations is administered by an institution. you can use that system to prove who you are but you cannot independently decide how the entire system operates.

calling this “the government owns your identity” would be inaccurate.

the real question is how much meaningful control you have over the collection and use of information about you.

and we should be precise about security.

the government says Aadhaar demographic data is encrypted both at rest and in transit. so saying we have no clue whether encryption exists would be wrong.

what citizens need is credible evidence that those protections work in practice. who can access the data? how are keys protected?

how quickly are failures discovered?

who is accountable when something goes wrong?

look at the recent eMudhra incident.

Cloudflare reported that private keys associated with eight intermediate certificates were compromised because the keys were weak. that creates opportunities for website impersonation and interception. it does not establish that Aadhaar's central database was breached. but it gives us a very serious reason to ask how security failures can survive inside institutions trusted to protect everyone else.

now suppose the Aadhaar database never suffers a breach. does that settle the privacy question?

no. a system can remain secure against outsiders while information is used in ways people never meaningfully understood or expected.

security asks who can get in. privacy also asks what those already inside are allowed to do.

if tomorrow an american AI company or an indian company seeks access to identity services or information connected to them through a government partnership then what exactly would that agreement permit? authentication? limited identity details? analysis of records held elsewhere? these are different things.

Aadhaar's sharing regulations prohibit sharing core biometrics. an agreement cannot simply be treated as permission to hand over everyone's fingerprints and iris data. and we should not present a hypothetical AI deal as something already happening.

the concern i see is bigger than a company wanting your fingerprint. it is the possibility of connecting a verified person to records across different parts of their life.

where such linking becomes possible and permitted it could make profiling far more powerful. those records do not have to sit inside Aadhaar itself for that risk to exist. a common identifier can become the bridge between databases.

this is why i think india's identity infrastructure could become enormously valuable in an AI economy.

verifying real people could matter for distributing benefits and limiting duplicate claims. but that value should not automatically become a licence to collect more information about everyone. proving that someone is eligible for a payment should not require exposing their entire life.

and public benefit delivery is already happening. by may 2025 the government reported cumulative DBT transfers exceeding ₹43.95 lakh crore.

we should build on that experience. UBI would still require funding and political decisions. an identity system can help administer a programme but it cannot create the money needed to sustain it.

we also have some foundations for giving people more control. UIDAI's offline e-KYC allows digitally signed identity information to be verified without revealing the Aadhaar number or requiring fingerprints or iris data for that verification.

UIDAI also says this sharing can happen without it knowing about each use. that is a useful direction to build on.

what i want next is identity that people can carry and use with much more specific permissions. prove only what the service needs. make access understandable. let people challenge misuse and correct errors. give them interoperable payment options and a practical recovery process. nobody should lose access to essential benefits because a biometric check fails or they cannot operate a wallet.

and putting identity on a public blockchain would not automatically solve this. permanently exposing personal records could make the problem worse. the goal should be verifiable claims with minimum

disclosure and enforceable rights around whatever information still needs to be collected.

i see the next two years as an important window to demand these choices before more systems become dependent on the same identity infrastructure. but privacy does not become a fundamental rights issue two years from now. the supreme court recognised it as a fundamental right in 2017. the responsibility already exists.

we have built infrastructure that lets institutions verify more than a billion people. now we need to give those people stronger power to question how institutions use that ability.

your identity should help you participate in the economy without making your entire life available to everyone who wants to build a business around it.

portable identity, onchain

Internet collectivism demands portable identities. Portable identities could be designed onchain when you can link both offchain and onchain data of human/agent behind the public key to a non fungible, fungible, semi fungible, diffusive or any kind of tokens you want to design, there are 50+ different ways to think about it.. build a platform where people could create those onchain portable identities with verifiable private metadata.

the aadhaar lesson and the onchain lesson point in the same direction. putting personal data on a public blockchain is not the goal, it would be a disaster. the goal is a public key you control, a human-readable name on top of it, and private claims you can prove without revealing them. verify the minimum. reveal by choice. carry it everywhere.

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agents with names

everything in this part applies to a new kind of citizen too. AI agents are going to hold money, sign agreements, and do work. they will need identity, privacy, and reputation even more urgently than we do, because nobody can look them in the eye.

NFTs could make their next comeback as identities for AI agents.

> an agent you can discover.

> an agent whose work you can inspect.

> an agent with a reputation that follows it across applications.

NFTs paired with tokens, new trading rules, and added utility are interesting.

but i still see those as early experiments.

the bigger idea starts when you think of NFTs as identities for software that actually does useful work.

and for CT, useful often means helping people find opportunities, save time, or avoid losing money.

imagine an AIXBT-style research agent with a public record attached to its identity.

it tracks wallets.

studies token launches.

flags changes in liquidity.

publishes research with sources.

but now you can look beyond its latest viral post.

what did it say before the move?

how often was it wrong?

did its results survive fees and slippage?

are people actually paying to use it?

that is where things get interesting.

there is a proposed Ethereum standard called ERC-8004, currently in draft.

it gives agents an NFT-based identity and provides registries for reputation feedback and independent validation.

the agent itself can still run offchain.

the onchain part gives applications a common way to identify it and inspect recorded signals about its work.

now imagine a marketplace where you compare agents by their actual service history.

one is good at researching new protocols.

another monitors treasury risks.

another finds execution routes.

another checks the first agent's work.

you could hire them based on evidence.

other agents could do the same.

a research agent hires a data agent.

an execution agent checks with a risk agent.

each service builds a history around a recognizable identity.

payments and task settlement would need separate infrastructure, but the identity gives those interactions a consistent reference.

the next layer could be businesses built around these agents.

paid research.

monitoring subscriptions.

specialized APIs.

automated services that customers keep coming back to.

but there is a question every NFT buyer should ask:

what exactly do i own?

the agent's registration?

access to its service?

control of its operation?

a contractual share of revenue?

those are different things.

buying an identity NFT does not automatically hand you the model, servers, customer relationships, or earnings.

and reputation needs scrutiny too.

fake accounts can manufacture positive feedback. an onchain rating is only useful when you understand who submitted it and what evidence supports it.

my thesis is that the market could wake up to this when an agent shows useful, repeatable results that people will pay for.

then we could see a new narrative around “8004 agents” or “onchain agents.”

the opportunity i’m watching is where a working product, credible performance history, and clearly defined ownership come together.

a low floor price alone tells me very little.

a cheap NFT can still lose its entire value, and an agent’s success might never flow to its holders.

but an agent with real customers, useful output, and meaningful rights attached to ownership?

that is worth researching.

the next NFT people care about could be the identity of a service they use every day.

the stack already exists

You can literally register onchain agent, that registration makes it more like an identity of agents, and than you link agents with ens name, and a PFP, PFP would be the persona of the agents, two agents can talk each other privately on XMTP, and they can move money using X402... this is all possible, and working.

the pieces in that list each do one job. an agent registry like ERC-8004 gives an agent a persistent identity and a place for reputation and validation signals. an ens name makes it findable. a pfp gives it a recognisable persona. a private messaging layer like XMTP lets agents talk. a payment protocol like x402 lets them pay for services over normal web requests. none of these is the whole thing. together they look like the beginnings of an agentic economy.

due diligence for the agent era

Nothing could be more speculative than onchain agents at this point in crypto.

You literally know that agents can't move money without being onchain.

Onchain agents is a mainstream thing.

Now all you need to figure out projects which are trying to create an ecosystem around onchain agents.

It's super easy when you understood what's possible.

All you need to have understanding on all key ERCs which got released in context to agents.

8004 is the one.

How many people actually know you can actually track data on onchain agents.

That's where you will clearly get to see projects with actual agents

Next step you figure agents role what they do what kind of productive outcome they are bringing

Are they fad or doing something real?

Once you get that data you could build conviction on token.

On token side buyback and burn is your meta + if the token is using custom hooks to make the token design more interesting

That's your due diligence.

You do and you would be early to so many projects by your own.

Use AI to simplify things which looks complex to you.

Understand write build your own audience

Now you ask where you could make money in all of this.

playing it out as narrative, talking more about agent NFTs, and how agent NFTs are front running the future.

You make money when you are early to those projects which are front running a possible future.

Also since now everyone can vibe code, you need to get early to first movers, because second movers would be in hundreds of them so attention would be highly diluted in that case also both first and second movers will attract many vulnerabilities, so don't be surprise some of them get hacked.

So all you get a chance to get early, put small capital, locate higher upside potential opportunities and get out when you see the first major wave when you find your whole feed talking about.

rewarding commitments instead of attention

and here is the idea that connects agents to the next part of the book.

Much of the internet economy rewards attention.

You create content.

You attract engagement.

You receive visibility.

You convert that visibility into economic value.

Now imagine an economy where rewards are connected to commitments.

You don't receive rewards simply because you promise something.

You receive them because you fulfill what you promised.

An agent accepts work.

It enters an agreement.

It delivers an outcome.

The result is verified.

The payment is released.

Its reputation improves.

Now imagine this happening across millions of agents and humans.

We could begin creating an economy where economic value is more closely connected to useful contributions.

Of course, not every task can be measured perfectly, and not every failure deserves punishment.

But the broader question is interesting.

What if the next generation of digital economies rewarded the fulfillment of commitments rather than the ability to manipulate attention?

§ part 6

part six

human commitment markets

putting something at stake to become better

if there is one idea in this book i would build tomorrow, it is in this part.

crypto has proven, beyond any doubt, that people will commit extraordinary amounts of capital to back their beliefs. the problem is what they are betting on: mostly prices, mostly other people's attention, mostly the hope that someone else buys later. what if the same instinct, the same thrill of backing your own judgment, could be pointed at things you can actually improve? your skills. your knowledge. your promises.

that's what i mean by human commitment markets. people commit something, code checks the outcome, and the reward goes to fulfilment, not to noise.

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what will you commit?

Crypto keeps asking the wrong question: **what is the next narrative?** Memes, perps, prediction markets, AI agents, tokenized stocks. Every few months, the market finds a new wrapper, pushes capital into it, and then moves on to the next thing.

Maybe the better question is: **what kind of economy are we actually trying to build here?**

Because if the final answer is simply another casino where people with more capital, more attention, more followers, better information, and faster execution extract money from everyone else, then we haven't really built anything new. We just made the old game faster and put it onchain.

If you look at the history of crypto, one thing keeps repeating: commitment.

In the early days, people committed personal computing power to mine Bitcoin. Then people committed GPUs to Ethereum. Later, crypto slowly moved toward a much simpler model: commit capital

and get capital. The more capital you had, the more opportunities you could access and the more returns you could potentially generate.

Then came another phase where people started committing labour. They farmed airdrops, joined Discords, used protocols, created content, built communities, completed tasks, and clicked buttons every day hoping that six months later someone would drop tokens into their wallets.

Sometimes it worked. Sometimes people made life-changing money. Sometimes greed made them round-trip those gains back into the market.

But the core mechanism stayed the same: you commit something and expect an outcome. That commitment could be time, capital, computing power, attention, reputation, or labour.

So maybe the next generation of crypto products should start with a very simple question: **what are people willing to commit, and what outcome can code fairly give them in return?**

Once you think this way, the design space becomes much larger than memes and perpetuals.

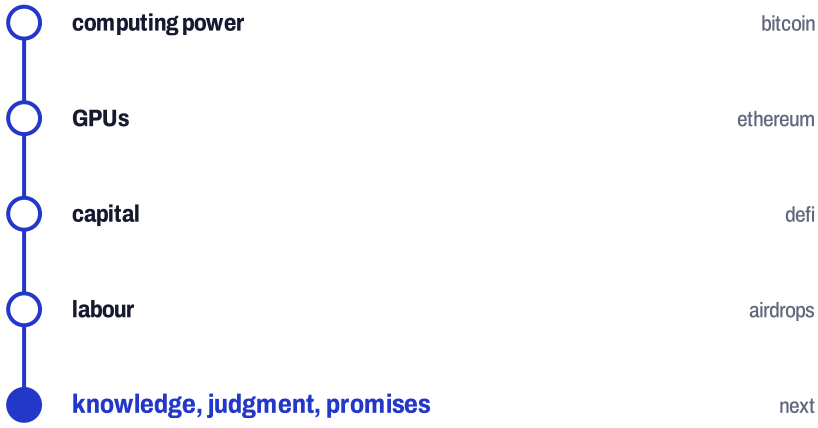


fig. 9 what people commit, era by era

why can't real-world actions become markets?

Imagine people pooling capital around an event. Not just elections or token prices, but an actual real-world event.

A marathon could become an event. A football tournament could become an event. A creator finishing a film, a community restoring a lake, a startup reaching a milestone, or a group of people completing a physical challenge could all become events around which capital and incentives are coordinated.

Suddenly crypto becomes something larger than trading assets. It starts becoming a system for coordinating people around outcomes.

This is where I think crypto becomes interesting again. If our end goal is mainstream adoption, why are we limiting ourselves to products where someone needs to become an extreme degen just to participate?

Why can't real-world actions be tokenized in ways where the risk is understandable and the upside is still exciting? Something closer to 1x understandable risk with the possibility of 2x or 3x outcomes feels much more mainstream than teaching everyone how to use 50x leverage.

The opportunity is not only to create new markets. It is to create new forms of participation.

three pillars

I keep coming back to three words: **commitment, coordination, and reputation.**

These might be three of the most important pillars of crypto, and they are still not properly synced with each other.

Crypto figured out capital. It figured out speculation. It figured out liquidity and token issuance. We became insanely good at creating assets, markets, and incentives. But humans still haven't figured out how to coordinate around rules without eventually corrupting those rules.

Maybe agents will.

If AI agents actually solve coordination at scale, I think a lot of it will happen onchain. Agents need rules, identities, payments, reputation systems, programmable environments, and ways to prove actions. They also need systems where they can coordinate without constantly asking a centralized operator for permission.

Crypto already has many of these building blocks. They are just scattered across different protocols and standards.

That is why the combination of AI agents and crypto is more interesting to me from a coordination perspective than from the usual "AI token" perspective.

commitment.

coordination.

reputation.

three pillars of crypto. (they should find sync with each other thats not the case now)

humans failed at achieving it through code.

i hope agents will achieve it.

if they do, they will do it onchain.

there is no other ways.

for now as humans what we can do best is to educate ourselves about what these 3 missing pillars mean for growth of cryptology for better society.

what makes a commitment market fair

let me add some design principles, because "bet on outcomes" can turn ugly very quickly if you are careless.

first, the outcome has to be verifiable by someone other than the person who profits from it. a marathon time from a timing chip is verifiable. "i worked hard" is not. second, the risk has to be understandable before you enter. the beauty of a 1x-risk, 2-3x-upside design is not the multiple, it is that a normal person can explain it to their family. third, the market should reward effort the participant controls, not luck or insider information. and fourth, losing should

leave you with something: a lesson, a skill, a reason to come back stronger. if losing only leaves you poorer, you built a casino with better marketing.

hold these four principles in mind for the next chapters. every design that follows is an attempt to meet them.

24

betting on yourself

I think there is a scope for creating a whole library of games where people could enter with a token ($ETH, $SOL, $ZEC, $NEAR), anything but stablecoins.. and compete with others to prove how AI literate they really are. Those who prove they are more literate than others take the winnings home.. the rest lose and try again. The whole idea here is to let 10% of people win something, while 90% lose.. and give 100% of them a reason to believe they have a chance of winning big by backing their own claim that they are more AI literate than others.

Now think about what we are really creating here.. we are creating a market where people are willing to commit their capital to prove their knowledge.. and unlike most crypto markets where people are betting on the future price of an asset, here they are betting on something they can actually improve through their own efforts. Their knowledge. And that's where things start getting interesting.. because when someone loses money in a speculative market, they might spend the next few hours looking for another token to recover their losses.. but when someone loses a competition because

another person understands AI better than them, they have a very different opportunity in front of them. They can learn, improve, and come back with better skills. And the more they learn, the better their chances of winning future competitions. So now we are introducing an economic incentive for people to become more AI literate.. something which I believe is going to be one of the most important skills for society in the coming decade.

But I don't want to stop at games and financial rewards.. I want to create a reputation economy around the whole thing. Every participant should have an identity.. and I think ENS could become an interesting layer here. You enter with your wallet, register or connect an ENS name, get a PFP, and start building your reputation through the competitions you participate in. Your ENS becomes your identity.. your PFP becomes your recognizable character.. and your reputation becomes a reflection of what you have actually demonstrated over time. Not how many followers you have, not how loudly you talk about AI, not how many AI tools you have listed in your bio.. but how well you can actually use AI to solve problems.

And the more competitions you participate in, the more data we collect about your strengths and weaknesses. Someone might be exceptionally good at identifying AI-generated images.. someone else might be better at using AI agents to complete complex tasks.. another person might understand hallucinations, reasoning failures, privacy risks, or the economics of AI much better than others. We should not treat all of them as equally skilled just because they won a few games. We should create different reputation categories around different forms of AI literacy. And that's where the leaderboard starts becoming something much more valuable than a leaderboard.. it becomes a discovery engine for human expertise.

Think about it.. we spend so much time discussing how AI will replace human labour.. but what if we could create a market which helps us discover humans who are exceptionally good at working with AI, and then use their expertise to make other humans better at it? Imagine someone from a small town in India participating in these games for six months.. winning multiple competitions, developing an excellent reputation, and eventually becoming one of the highest-ranked participants in their region. Now that person has something which is difficult to fake.. a history of demonstrated skills. And what if we could use that history to identify people who could become AI literacy trainers in their own communities?

They could teach school students, college students, teachers, small business owners, farmers, local entrepreneurs, and even older people who have never used an AI tool before. And we could reward those trainers from the treasury.. not by printing another token, not by promising some future governance rights, but from the actual fees the platform earns through the competitions. So the more people participate in these games, the more money the platform generates.. and a portion of that money goes back into making society more AI literate. And this is where I see a beautiful economic loop emerging.. people commit capital to prove their skills, competitions generate fees, skilled people build reputation, reputation helps us discover potential educators, educators take AI literacy to offline communities, and some of those newly educated people eventually become participants, competitors, winners, and educators themselves. The whole system starts feeding itself.

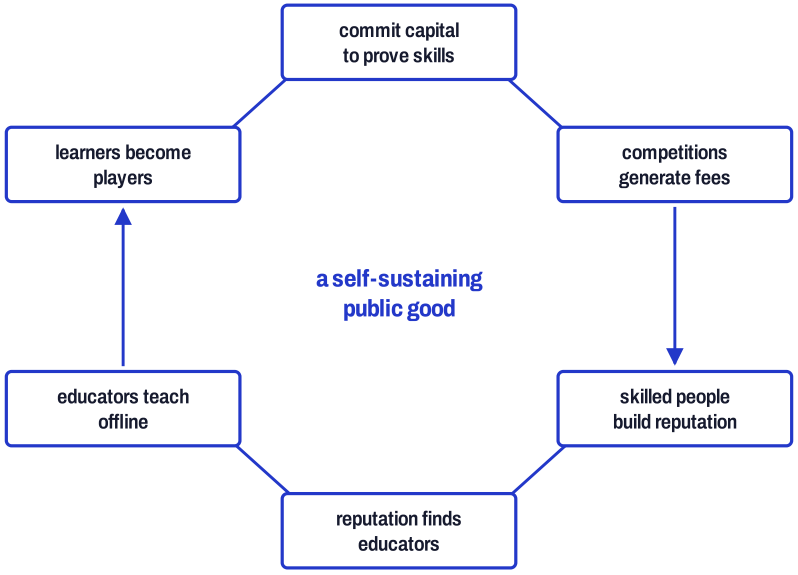


fig. 10 the AI literacy loop: the system starts feeding itself

Now let's talk about the games.. I think there is enormous scope for designing simple games which make people understand AI without making them feel like they are attending another boring online course. For example.. can you tell whether something was generated by AI or a human? You are shown two images.. one is AI-generated and the other is human-created.. and you have to identify which one is which. But instead of simply guessing, you should also explain what made you reach that conclusion. And this matters because as AI-generated content becomes increasingly difficult to distinguish from human-created content, people will need to develop better habits of verification rather than relying on visual intuition. We could have similar games around audio, videos, news articles, research papers, advertisements, and social media posts.

But here's another interesting possibility.. what if the game is not about identifying AI-generated content but identifying which AI-generated answer is actually correct? Imagine five different AI models answering the same question.. four give convincing but incorrect answers, and one gives a properly supported answer. Your task is to identify the most reliable answer and explain why. Now we are testing something much more important than someone's ability to write prompts.. we are testing their judgment. And judgment is going to become increasingly valuable in a world where generating answers becomes almost free.

We could create games around prompt engineering.. where participants receive the same problem, access to the same AI model, and limited time or compute credits to produce the best outcome. We could create agent competitions where people have to build agents that complete real-world tasks.. and the agents are evaluated on accuracy, cost, efficiency, safety, and actual results. We could create hallucination-hunting games where people compete to identify factual mistakes in AI-generated research. We could create privacy games where participants must identify what sensitive information an AI application is collecting or leaking. We could even create games where participants have to decide when not to use AI.. because being AI literate should not mean blindly using AI for everything. It should mean understanding what AI can do, what it cannot do, when to trust it, when to question it, and when human judgment should take priority.

Someone who can generate 100 images using AI is not necessarily more AI literate than someone who can identify a manipulated image being used to spread misinformation. Someone who can build an agent in five minutes is not necessarily more AI literate than someone who understands why that agent should not have unrestricted access to someone's money. So we need to think about

AI literacy as a combination of technical ability, critical thinking, creativity, verification, ethics, and practical problem-solving.. and design different games around each of these abilities.

Now imagine we have hundreds of such games.. some take 30 seconds, some take five minutes, some require teams, some involve building real products, some involve identifying misinformation, some are purely educational, and some are advanced enough that even experienced AI engineers would find them challenging. Every game produces evidence about what a participant knows.. and over time, the reputation system becomes more meaningful. But we need to be careful about how we design the reputation layer.. because if someone can simply create ten wallets and participate using ten identities, the leaderboard becomes meaningless. If someone can use an AI agent secretly to solve a challenge which is supposed to test their own knowledge, we are no longer measuring what we think we are measuring. And if someone can buy their way into the top rankings through repeated paid entries, we are simply recreating another capital-dominated system.

So reputation should not be based on how much money someone has won.. it should be based on how consistently they have demonstrated skills, the difficulty of the challenges they completed, the quality of their explanations, their performance against other verified participants, and how their abilities have improved over time. We could use ENS as the public identity layer.. and use privacy-preserving verification mechanisms for those who want to establish that they are unique humans without revealing unnecessary personal information. We should also separate reputation earned through independent human reasoning from reputation earned through AI-assisted work.. both are valuable, but they measure different abilities.

And when someone becomes an educator, we should introduce another reputation layer around their teaching outcomes.. because being good at something and being good at teaching it are two different things. Imagine a highly ranked player organizing an offline AI literacy workshop for 50 students. The treasury sponsors the workshop.. but the reward should not depend only on the number of students attending. We should measure how much those students actually learned. Give them a simple assessment before the workshop.. another after it.. and perhaps another a few weeks later. Now we have evidence that the educator actually helped people improve. And that contribution becomes part of their reputation too. So reputation is no longer just about winning games.. it is also about how much knowledge you helped distribute.

Now there is another thing I want to explore.. why should we limit participation to one token? Someone holding ETH should be able to enter using ETH.. someone holding SOL should be able to enter using SOL.. someone holding ZEC or NEAR should have the same opportunity. I am deliberately thinking about non-stablecoin assets here because I want to explore how existing crypto communities could participate in a shared market of knowledge and skill without needing another token. Of course, we would need to design how entry fees and prizes are accounted for when the underlying assets have different prices and volatility. We could have separate token-denominated competitions, or use transparent reference prices to create comparable entry tiers without requiring people to deposit stablecoins. The treasury would also need clear rules for managing volatility and funding commitments. And the important thing is that the token someone holds should not decide how much reputation they can earn.. capital should determine how much someone is willing to risk, not how intelligent or skilled the system considers them.

Now about the 10% winners and 90% losers model.. I think this creates an interesting competitive structure, but we need to distinguish between financial outcomes and educational outcomes. Financially, 10% might receive prizes while 90% receive nothing from the prize pool. But educationally, we should design the system so that even the people who lose have an opportunity to gain something meaningful. They should know why they lost, which skills they need to improve, and how they can practice without repeatedly putting money at risk. Otherwise we will end up creating another speculative market which uses education as a narrative to encourage people to lose money. And that is not the kind of economy I want to build. The competitive element should create excitement.. but the learning element should create lasting value. We would also need to handle paid-entry competitions carefully, because prize pools funded by losing participants can raise gambling and skill-contest regulatory questions depending on the jurisdiction. Free-to-play challenges, transparent fees, spending limits, anti-collusion systems, and independently evaluated skill-based outcomes should be part of the design from the beginning.

Now imagine the scale of this.. a thousand people participate in competitions every day, then ten thousand, then a hundred thousand. Some people come because they want to win, some because they want to prove themselves, some because they want to improve their skills, some because they want to build a reputation, and some eventually discover that they can make a living teaching other people what they have learned. We are essentially using market incentives to create a distributed network of AI literacy educators. And the most interesting part is that we don't necessarily need universities, corporations, or governments to tell us who is qualified. We can use verifiable performance, reputation, peer evaluation, and demonstrated teaching outcomes to discover people who can actually contribute. Of course, this would not replace

formal education or professional credentials everywhere.. but it could create an alternative pathway for people who have the skills without having the degrees.

And if we get the treasury economics right, the system could finance its own social impact rather than depending entirely on donations. For example, imagine 10,000 entries in a day with an average entry value of $5.. that's $50,000 in daily entry volume. If the platform charges a transparent 10% fee, that's $5,000 in gross daily revenue before operating costs. If we allocate 30% of those fees to AI literacy initiatives, that's $1,500 a day.. or roughly $45,000 over 30 days at the same activity level. And that money could sponsor local workshops, community educators, learning materials, and access to AI tools for people who otherwise could not afford them. These are just illustrative numbers.. actual participation, repeat demand, fees, operating costs, and legal restrictions would determine whether such a model is sustainable.

But you can see what I am trying to explore here.. what if the same human desire to compete, prove superiority, and win financial rewards could be redirected toward building a more educated society? Crypto has already demonstrated that people are willing to commit extraordinary amounts of capital to their beliefs. They buy tokens because they believe they understand the future better than others. They trade because they believe their judgment is superior. They participate in prediction markets because they believe they know something others don't. But most of these markets reward being right about prices or outcomes. What if we create markets which reward becoming more capable? What if instead of simply betting on which AI company will dominate the future, people could compete to prove they understand AI better than others? And what if the financial activity generated by those competitions could fund the education of people who are currently being left behind by the AI revolution?

I think there is a much bigger idea hidden here than simply creating another gaming platform.. we could be creating an economy where knowledge becomes something people are willing to back with capital, reputation becomes evidence of demonstrated ability, and a portion of the value generated by competition is used to distribute that knowledge across society. And perhaps this is one of the directions crypto should explore more seriously. Not another token, not another narrative, not another market where attention is the only thing being traded.. but markets where people have an incentive to become better at something, and where their improvement can eventually become valuable to people beyond themselves.

Because the real winner of an AI literacy game should not just be the person who takes the prize home.. it should also be the society which becomes more capable because that competition existed. And if we could make learning as exciting as speculation.. while ensuring that people are rewarded for genuine skills rather than simply encouraged to risk more money.. we might end up building something much more meaningful than another crypto game. We might end up building a market which turns human curiosity, competition, and knowledge into a self-sustaining public good.

judgment is the new scarce skill

In such challenges when majority have false sense of judgment, minority will always win more returns on the commited capital they put to back their judgement.

Minority always wins in context to proving your worth publicly.

Majority always lose. Thats the normal rule in the economy.

But also those who lose will get a chance to win given if they could improve their sense of judgement by constant practice, consuming knowledge.

Think about having a casino for human society where both knowledge lead to make more money , and seeing more people making money lead humans to acquire more knowledge.

judgment of thoughts and actions become so valuable in human society in the future where thoughts and actions would be driven mainly by agents.

The better you could judge, and prove your history of judgement publicly the higher reputation you get in the society.

Higher reputation eventually leads to valuable relationships.

Valuable relationship creates value in terms of both knowledge and money.

The whole movement of human judgments over ai outputs could happen onchain in a way where better judgment is being rewarded not by creating new tokens but by creating events where people could participate to bet on their own judgement.

This is much bigger than what we have seen in crypto so far.

> *what if we create markets which reward becoming more capable?*

25

proof of skill

the AI literacy games are one example of a much bigger pattern. this chapter widens the lens: an entire labour market where your demonstrated outcomes are your qualification.

the next big crypto economy might not need a token.. it needs proof of human intelligence amplified by AI..

an advanced AI user can offer services to a basic AI user and help them achieve outcomes they otherwise couldn't achieve on their own.. the difference between these two people is not access to AI but how effectively they can use it..

now imagine building an entire economy around this difference.. where people don't need to launch tokens to attract speculation but instead commit capital to prove their skills..

different challenges could be created where people compete to demonstrate how much better outcomes they can produce using AI.. coding, research, filmmaking, design, marketing, problem-solving or even building entire businesses..

participants stake capital to enter these challenges.. anyone scoring 60% or above gets rewarded, while those below 60% lose part of their stake.. the slashed capital gets distributed among successful participants, with higher scores receiving larger rewards..

but the interesting part begins after the competition..

every challenge creates verifiable proof of skills.. every successful outcome builds reputation.. buyers can discover skilled people based on their actual performance rather than degrees, resumes or social media followers..

imagine a decentralized labour market where your reputation becomes your qualification, your past outcomes become your portfolio, and your ability to solve problems determines your earning potential..

of course, challenges need independent evaluation, anti-collusion mechanisms and safeguards against capital-based advantages..

this could eventually transform education, qualification and hiring.. people would learn, compete, earn and get hired within the same ecosystem..

crypto could become the coordination and settlement layer of this new economy.. p2p, decentralized and distributed..

we spent years creating markets to speculate on tokens.. perhaps it's time to create markets where people commit capital to prove what they can actually do.

learn, then earn, and keep what you learned

remember the airdrop farmer who felt happy with a hundred dollars until he saw someone else make ten thousand? this is where i want to take that story.

but this time what you learn should stay useful even when the incentives stop. you should be able to research and write. build and teach. explain something difficult to someone who doesn't understand it. because if all you learned was how to qualify for the next payout then we just created another version of the same loop.

incentives could be given in synthetic tokenized stocks or even $ETH (i believe ETH is better private money than ZEC, and most of you will agree in next 12 months)

@VitalikButerin @RAILGUN_Project @ethereum i'd love to explore something where people learn about privacy tech and core cypherpunk ethos and the philosophy behind crypto..not just shilling a token but understanding why privacy matters. why permissionless money was worth building. what these ideas mean in their own lives. i know you talk about this very well but you're a busy man. and i want to give my time to teaching those who really care.

there are thousands of unproductive human forces doing nothing but clicking buttons in hope of airdrops right now.

most of them are simply copy-pasting steps.

more like a "generational opportunity to change lives forever" wasted.

what if airdrop farmers of the past can really get involved in something which could give them job and business opportunities for the long term.

all it needs is effort and hard work.

a commitment to learn.

and if they fulfill their commitments, they'll make money.

if they break their commitments, they'll lose money.

better than spending money on school and college, where you don't need to pay any fees, rather you will earn fees if you pass exams.

imagine a future where students get money to complete a course, not like the current model where you pay money, learn nothing, and become an unproductive job force.

crypto was supposed to make people think freely, explore without fear, shame, and validation.. we got tools for it.

AI is supposed to make the whole cycle of learning to creating tangible value fast.

we got two superpowers which could lift the poor and make them rich, given they get the right guidance.

all we need are people who really want to grind.

and human commitment markets for them only.

creators deserve a fix too

content creation is largely democratized but the core problem is the fair distribution of value for people who create good content for the joy of it, they show a constant effort, become delusional optimistic for few days, weeks, months, and years that varies to series of conflicts (emotional, mental, physical) in every moment we're spending in the lap of mother earth who herself is so wounded right now, its becoming hard for her to feed you....(people who could get this message could align with how bad content on the internet have became)..and crypto could offer a fix when you create a human commitment markets.

build a community which learns to create content using AI together.. not slops but genius content.

number of people who could create genius outcomes using AI are the rare breeds if they can keep the momentum for 18 months than they can surely attract massive pie of rewards in those times when most of the workforce of the world will lose trust in any kind of fair incentive from the economy they have been part of.

In those times.. crypto will find its meaning again in the society again.

26

why is my phone so hot?

this is the longest single idea in the book, and i've kept it nearly intact, because it shows how all the threads connect in one concrete experiment: data, privacy, identity, reputation, commitments, education, and a token that begins without a market.

what if we could turn millions of smartphones into a community-owned research network, reward people for contributing data, and use that network to experiment with an entirely new kind of token economy..

your phone is working even when you are not

have you ever wondered why your phone is getting hotter even when you are not doing anything particularly complicated.. you are not mining bitcoin.. you are not running some massive ai model locally.. you are probably just scrolling instagram, watching videos, using chatgpt, taking photographs, or switching between applications.. and yet your phone is heating up, your battery is draining, and every few weeks you are deleting photographs

because your storage is running out.. something interesting is happening here.. we are carrying increasingly powerful computers in our pockets, but most of us have almost no idea how their computing resources are being used.. think about a normal smartphone user who has 50–60 applications installed.. some applications are actively running.. others are receiving notifications, syncing information, downloading content, processing photographs, or performing background activities whenever the operating system allows them..

every application needs some combination of computing power, memory, storage, network connectivity, and energy.. but how much.. how frequently.. and how much of that activity is actually necessary.. there is an important distinction here.. having 60 applications installed does not mean 60 applications are continuously running.. modern operating systems like ios suspend most inactive applications and carefully restrict background execution.. still, the broader question remains unanswered for most users.. who is consuming the computing resources of our phones, and how much control do we really have over that consumption.. i think this question becomes much more interesting when we introduce ai into the equation..

ai is changing our relationship with personal computing

for decades, we have been upgrading our phones for better cameras, faster processors, larger screens, and more storage.. now ai is becoming another reason to upgrade.. but there are two different kinds of ai activity happening on smartphones.. one involves running models directly on the device, where the phone's cpu, gpu, or neural processing hardware performs the inference.. the other involves sending requests to cloud-based models, where much of the computation happens in remote data centers.. these are very

different workloads.. a cloud-based ai conversation might require relatively little local computation compared with running a large model directly on the phone.. but ai applications can still consume energy through networking, rendering, audio processing, image manipulation, and other activities..

now imagine combining these workloads with gaming, video streaming, charging, navigation, background synchronization, and direct sunlight.. your phone has a limited thermal budget.. it cannot continuously consume unlimited power without producing heat.. and that brings me to something i find particularly interesting.. what happens when we start measuring smartphone usage not just in hours, but in the computing resources and thermal pressure associated with those hours..

what does an eight-hour smartphone user actually consume

let's divide smartphone users into five groups based on daily screen time.. 2 hours: messaging, calls, occasional browsing.. 4 hours: social media, browsing, videos.. 6 hours: heavy social media, streaming, multitasking.. 8 hours: extended media consumption, gaming, ai tools.. 10+ hours: near-continuous mixed smartphone activity.. these are hypothetical profiles, not measurements of actual computing demand.. an eight-hour user watching lightweight videos might consume less peak processing power than a two-hour user playing a demanding game.. screen time alone cannot tell us how much computing power is being consumed.. so we need additional variables.. we need to understand the device model, battery condition, charging behavior, storage pressure, operating-system version, thermal state, network conditions, and the types of activities users perform.. and perhaps we need to create a new measurement.. something that tells ordinary people how much

pressure their daily digital behavior is putting on their devices.. not simply how many hours they are spending online.. but how efficiently their phones are operating..

could your phone become dangerous

imagine you are running a demanding application on your phone.. you leave it charging near a window under direct sunlight.. the phone starts heating up.. its operating system detects thermal pressure and begins reducing performance.. but you are not there.. what happens next.. modern iphones have several thermal protections, including performance throttling, restrictions on demanding activities, and temperature warnings.. apple's thermal-state system distinguishes between normal, elevated, serious, and critical conditions.. these protections substantially reduce the risk of overheating.. a hot phone does not automatically mean an explosion is likely.. smartphone battery fires are uncommon, and serious incidents are more closely associated with conditions such as damaged or defective batteries, physical damage, or unsafe charging equipment..

but that does not mean thermal safety is unimportant.. a phone left in direct sunlight, exposed to extreme heat, or used with a damaged battery deserves attention.. instead of frightening people with exaggerated claims, why don't we help them understand the conditions that make their devices less efficient or potentially unsafe.. this could become a useful public education initiative.. and perhaps an interesting open research project..

what if we built a lightweight application that measures smartphone health

imagine installing a small application that helps you understand the operating conditions of your phone.. it does not read your private messages.. it does not collect your photographs.. it does not track your personal conversations.. instead, it uses whatever device-health signals the operating system legitimately makes available, combines them with optional user-provided information, and creates a transparent risk profile.. something like this: your phone's thermal state, battery signals, and usage habits feed into a risk engine, which produces your score and a research contribution.. we could create an experimental score between 0 and 100.. a lower score could represent fewer observed warning signals, while a higher score could indicate conditions worth investigating.. but this would need careful design..

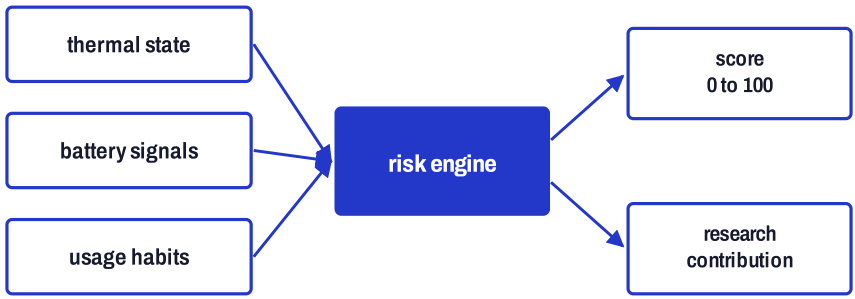


fig. 11 from phone signals to a score and a research contribution

an ordinary ios application cannot access every other application's cpu usage, obtain unrestricted background activity logs, or read the phone's internal temperature sensors directly.. it can access certain thermal-state information for its own process, while broader device

information is limited by apple's privacy and security architecture.. so our first task would be to build a scientifically defensible score using accessible measurements and clearly distinguish observed data from estimates.. the score should never pretend to predict the probability of a battery explosion.. its purpose would be to encourage better device usage, identify recurring thermal patterns, and generate useful research data.. and this is where the idea becomes much bigger..

what if people owned the network that collected this data

today, enormous amounts of information are generated through smartphones.. technology companies use different categories of data to improve products, personalize experiences, diagnose problems, and optimize their systems.. but imagine a community-owned research network where people voluntarily contribute limited, privacy-preserving information about how their devices behave.. we could begin with 1,000 people.. then 10,000.. eventually perhaps a million.. different devices.. different climates.. different usage patterns.. different battery ages.. different operating systems.. think about what we could learn.. do older phones experience serious thermal states more frequently.. how does charging behavior interact with demanding applications.. are there meaningful differences between users in hot climates and cooler regions.. what happens to device performance when storage is almost full..

how do ai-heavy workflows compare with traditional social-media consumption.. we could publish research reports using aggregated data and allow independent researchers to examine our methodology.. and people contributing to this network should receive recognition.. perhaps that recognition could become an on-

chain token.. not because we need another speculative asset.. but because we want to experiment with how contribution, education, and collective decision-making could become part of token design..

a token that begins without a market

this is where i want to challenge how crypto projects are usually created.. most projects begin with a token design, a supply schedule, an allocation table, and eventually a market.. then everyone starts talking about price.. what if we reverse the process.. we launch a simple erc-20 token, but initially keep it nontransferable or otherwise restricted from trading.. people earn it for verifiable contributions to the research network.. they do not buy it from an exchange.. they do not receive it for generating meaningless activity.. they earn it by helping us understand a real-world problem.. the token initially represents participation, not an investment or a promise of future profit.. and we do not need to decide every future utility on day one..

instead, the community can collectively study different token designs and decide which functions actually make sense.. perhaps some contributions should be represented through nfts.. perhaps reputation should remain nontransferable.. perhaps different forms of participation should receive different credentials.. perhaps the community eventually decides that a transferable token is unnecessary.. that should be a legitimate outcome too.. the important thing is that the token's purpose emerges from actual contribution rather than speculation.. we could also explore uniswap v4 hooks as an educational tool for designing conditional liquidity mechanisms.. hooks can customize aspects of pool behavior, but they cannot automatically impose arbitrary rules on every erc-20 transfer.. if we want transfer restrictions or conditional balances, those rules need to be implemented through suitable token

contracts, permissions, or other architecture.. and before any market is introduced, the community should understand what those mechanisms actually do..

what if the community learns token design by participating in economic experiments

here is another idea.. we create short educational challenges where people learn how to design better crypto economies.. instead of asking them to watch hours of lectures, we create three-day challenges.. each challenge focuses on one topic.. maybe identity.. maybe reputation.. maybe sybil resistance.. maybe erc-20 extensions.. maybe uniswap v4 hooks.. maybe how to design incentives that reduce exploitative, player-versus-player behavior.. and we introduce financial commitments.. imagine 1,000 people joining a challenge.. each person commits $10.. the total pool becomes $10,000.. participants have three days to complete the course and demonstrate what they learned.. those who successfully complete the challenge receive their rewards from the pool.. but how many people are likely to finish.. historical research into 221 massive open online courses found a median completion rate of approximately 12.6%.. another study of india's swayam courses reported an average completion rate of around 4.6%..

these figures are useful references, but they are not reliable predictions for a short, paid, crypto-focused challenge.. people who voluntarily stake money and have only three days to complete a course might behave very differently.. let's model several possibilities.. at 10% completion, 100 winners receive $100 each.. at 20%, 200 winners receive $50 each.. at 30%, 300 winners receive $33.33 each.. at 40%, 400 winners receive $25 each.. at 50%, 500 winners receive $20 each.. at 60%, 600 winners receive $16.67 each.. these are gross payouts per winner.. these figures assume the entire

$10,000 pool is distributed equally among successful participants, without fees or reserves.. at a 20% completion rate, 200 participants would receive $50 each, five times their initial commitment..

completion	winners	payout each	vs $10 stake
10%	100	$100.00	10x
20%	200	$50.00	5x
30%	300	$33.33	3.3x
40%	400	$25.00	2.5x
50%	500	$20.00	2x
60%	600	$16.67	1.7x

fig. 12 1,000 people, $10 each: gross payouts if the whole $10,000 pool goes to finishers

at a 50% completion rate, 500 participants would receive $20 each, twice their initial commitment.. beyond 50%, it becomes mathematically impossible to double everyone's stake using only the original pool.. of course, this introduces difficult questions.. should the people who fail to complete the course lose their entire commitment.. would such a system encourage learning or simply attract speculation.. could it become a form of gambling.. would participants rush through educational content just to collect rewards.. these are not small concerns.. a better version might return successful participants' deposits and distribute a limited bonus funded separately, while directing forfeitures toward a transparent education treasury, subject to legal review.. we could experiment with different structures and measure actual learning

outcomes.. the objective should not be to manufacture losers.. the objective should be to discover whether financial commitments can help people develop knowledge and complete meaningful tasks..

identity, privacy, and reputation could become the foundation

any system that distributes rewards attracts people who try to manipulate it.. imagine someone creating 100 wallets and pretending to be 100 different participants.. or using ai agents to complete assessments automatically.. or creating multiple identities to claim rewards repeatedly.. if we simply distribute tokens based on wallet addresses, sybil attacks become an obvious problem.. so we need identity and reputation.. i think ens could provide an interesting public identity layer.. people could participate through an ens name and build a history of verified contributions.. but ens alone does not prove that a wallet belongs to a unique human.. we would need additional verification methods.. potentially privacy-preserving credentials, proof-of-personhood systems, selective disclosure, randomized assessments, and verifiable completion records..

we could create a structure like this: a human participant uses an ens identity.. private data supports verification, while a public profile carries reputation.. verification and reputation feed into proof of learning, which determines reward eligibility.. the important distinction is between proving something and revealing everything.. a participant should be able to prove eligibility without publishing their real name, phone details, or private activity history on a blockchain.. the network should collect as little personal information as possible.. and reputation should reflect genuine contributions, not simply how many tokens someone owns..

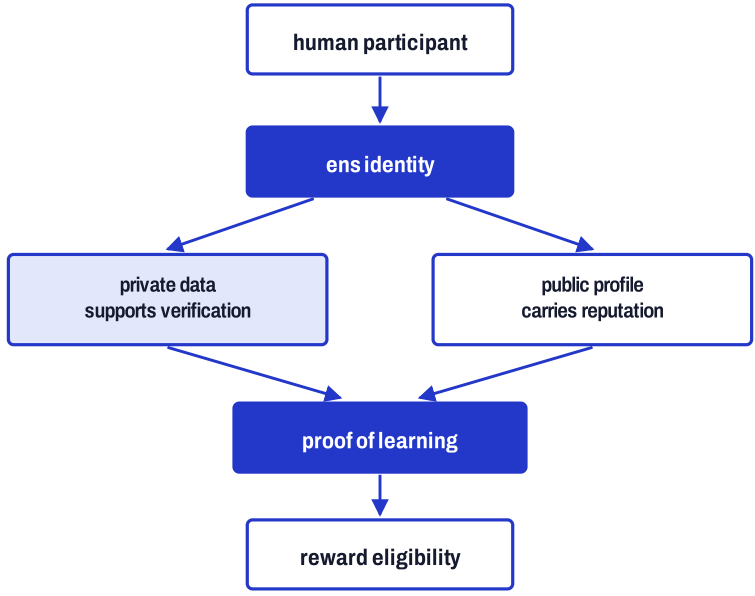


fig. 13 prove something without revealing everything

could we build a dao that designs its own token economy

imagine that the first 1,000 participants are not just users.. they are also the first researchers, learners, contributors, and potential governance participants.. we begin with a simple research application.. then we introduce contribution credentials.. then educational challenges.. then community discussions.. then governance experiments.. over time, the community could evaluate different token standards, governance mechanisms, and incentive models.. but we should avoid making every decision a token-weighted vote.. otherwise, we risk reproducing one of crypto's oldest problems: people with the most capital receive the most influence, regardless of their contributions.. instead, we could experiment with governance based on verified work, domain

expertise, reputation, and delegated responsibility.. an independent panel of researchers, developers, security experts, educators, and community representatives could review proposals..

we could use ai to help people understand proposed changes before voting.. and every significant change should be tested before deployment.. for example, the community might propose a new mechanism that rewards people for improving the accuracy of device-health observations.. or a system that funds independent research into smartphone energy consumption.. or an educational reward mechanism that cannot be exploited through repeated wallet creation.. these proposals could become practical exercises for people learning token engineering.. the dao would not merely govern an existing token.. it would become a laboratory for discovering what a token should actually be..

the bigger opportunity is not smartphone monitoring

the smartphone application is simply where the experiment begins.. the larger question is whether we can create a digital society where people voluntarily contribute information, learn new skills, establish verifiable reputations, and collectively design the rules of an economy.. there is an interesting historical parallel here.. before bitcoin, millions of people participated in peer-to-peer networks like bittorrent.. people shared bandwidth and storage, often without direct financial compensation.. those networks demonstrated that voluntary digital coordination could work at enormous scale.. bitcoin introduced another powerful idea: people could contribute computing resources to a network secured through economic incentives.. but somewhere along the way, much of crypto's public conversation became dominated by token prices.. we started measuring the success of networks by their market capitalization rather than the quality of coordination they enabled..

i think there is still an enormous opportunity to revisit the original promise of these systems.. not everything needs to begin with a liquid token.. not every participant needs to become a trader.. not every experiment needs a speculative market.. sometimes the most interesting thing we can do is bring a group of people together and attempt something that initially looks impossible.. what if we could create a network where a million people voluntarily contribute information about their smartphones.. what if that network produces genuinely useful public research.. what if participants learn how cryptographic identity, privacy, reputation, and economic commitments work.. and what if the community itself decides which token mechanisms deserve to exist.. maybe we succeed..

maybe the first token design fails.. maybe we discover that some incentives work better without tokens.. but the knowledge, relationships, research, and open-source infrastructure would still exist.. and that is what makes this experiment interesting to me.. we would not be creating another token and then searching for a reason for people to use it.. we would be creating a reason for people to coordinate, and then discovering whether a token could help.. perhaps the next interesting crypto experiment doesn't begin with a whitepaper, a trading pair, or a token launch.. perhaps it begins with a simple question: why is my phone getting so hot.. an experimental proposal by hitesh.eth.. the device-health scores, reward mechanisms, and governance architecture described here are concepts for research and testing, not deployed systems or established safety measurements..

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who builds the markets

do you remember that scene from ready player one where everyone is fighting for control of the oasis while IOI wants to turn the entire virtual world into another machine for extracting money from its users?

i think the next crypto war will look something similar..

we are so busy defending our tokens that we are barely questioning who will build the markets where the next billion people will participate..

stablecoins could bring billions of people into dollar-based economies.. tokenized stocks could let them trade american companies.. pre-ipo markets could eventually turn viral product launches into speculative opportunities..

imagine a startup releasing a banger ai video with a ticker below it.. the video becomes the new whitepaper.. attention attracts liquidity and liquidity creates valuations..

sounds familiar?.. we have seen this during the ico era..

but what if we use the same infrastructure to build something completely different?

imagine a person sitting in a village with nothing but a smartphone.. contributing to ai training, verifying data, completing useful tasks, building reputation and earning money directly from global markets..

imagine ai agents hiring humans, humans verifying agents and smart contracts distributing rewards based on actual outcomes..

now imagine millions of people participating in those markets using privacy-preserving identities and native crypto assets as collateral, settlement instruments and commitments..

that's where the real economic opportunity could emerge..

because whoever builds the markets will have enormous influence over how people earn, contribute and coordinate..

and if we leave that opportunity entirely to the same institutions we once wanted to challenge, we might end up becoming participants in their economy rather than creators of our own.

the real war

i'll connect this back to where the book began. in the first part i said the matrix doesn't need to destroy crypto, it can simply absorb it. in the third part i said we are in a second crypto war. this is what that war is actually about.

it is not about which token outperforms. it is about who designs the markets where the next billion people earn, learn, and coordinate. if those markets are built by the institutions we wanted to replace, people will participate on their terms: dollar-denominated, permissioned, surveilled, with value flowing up. if they are built by

us, they can be privacy-preserving, reputation-based, open to anyone with a phone, and paid in assets the community actually owns.

the tools are all in the previous chapters. purpose bound money. refundable and expiring tokens. portable identity. confidential balances. agent registries. commitment markets. proof of skill. nothing in this book requires a breakthrough that does not exist yet. it requires builders who would rather design markets than chase tickers.

§ part 7

part seven

the person behind the key

identity starts inside

we spent a whole part on digital identity. this short closing part is about the other kind. the one you build long before you ever generate a private key, and the one that decides whether any of these tools make your life better.

§ chapter 28

28

the persona

most of us are spending our entire lives trying to belong somewhere.

we want to be part of a collective where people share similar values. similar interests. similar emotions. somewhere we can sit with another human being and have a meaningful conversation.

but there is a hidden trade off in this whole process.

in the quest of belonging we start modifying ourselves.

we observe people before connecting with them. we try to understand what they like. what they value. what kind of personality they appreciate. and then we start building a version of ourselves which might be acceptable to them.

that is how the persona is born.

and almost everyone in society is doing this.

we are all wearing different masks to receive appreciation. validation. opportunities. relationships. money. status.

but the tragedy is that this whole process starts much earlier than we realize.

your parents were also maintaining their own persona in society. they wanted to be respected. they wanted to be seen as successful people. and when you were born you became an extension of their image.

your education became their social achievement. your career became their validation. your success became their status.

you might have wanted to become a musician or a cricketer but they wanted you to become an engineer.

and you were trained to believe that obedience was discipline.

school continued the same process. college continued it. workplaces continued it.

eventually you became a product of society.

a product which knows how to perform but has forgotten how to express.

and somewhere beneath that carefully maintained persona there is still a child who wanted to explore life differently.

that child never disappeared.

you simply stopped listening.

and the greatest tragedy is not that society failed to understand you.

it is that you spent your entire life trying to make society understand a person who was never truly you.

what if everything is data?

What if we look at every living being as an object carrying data?

You are an object with an identity, and that identity is connected to everything you have experienced since you were born.

Your memories, relationships, failures, achievements, habits, and decisions are all part of the data that shapes who you are today.

Now apply the same idea to different biological systems.

Your gut bacteria respond to information from their environment. Your neurons exchange signals. Animals continuously process information to survive.

Humans do the same, although our intelligence allows us to create much more complex interpretations of reality.

Now we are building AI systems that learn patterns from enormous amounts of data.

Different systems, different levels of intelligence, but information remains central to how they function.

And if information plays such an important role in survival and decision-making, shouldn't we question why its distribution is controlled by such a small number of powerful entities?

the artists we lost

most of the content creators amplify the same information at different flavors which you have consumed couple of times already.. don't you think it's very bad flow of information symmetry where the whole outcome is leading to growing concern, and fear during the time like today where all we see concerns all around us, people are losing hope.. give them hope but also make sure they know the truth.. but do you really think content creators of modern times are operating like the heroes of the past who used to shape society by content (poems, movies, songs basically all kind of performing and

creative arts).. content creators are artists.. do you really think majority of content creators are doing any kind of art to shape society these days?... the answer is NO.. when you realize that than you may think about fixing it even it could be a small effort you make to support those who speaking their mind out whenever they see information gaps in the society which is losing its privacy, data, reputation, identity ... hence trust.

I am just looking forward for the day thinking what value I could provide to the space than I just share my thoughts thinking one of them will resonate with many most of the people want to hear about their bags though which makes total sense to me than I think what to talk about their bags.. same old thing the max I could do optimize hope more or share some insights which was unheard by many.. the second option makes more sense to me as that would help me

Learn something myself which I could share with people who still think they could learn something new everyday on CT

i'll be honest about why i write. most days, people want to hear about their bags, and i understand that. but i would rather share one thought that someone hasn't heard before than repeat the same thing a hundred accounts already said. if a creator's job is to shape society, then the least we can do is try.

for those who stay

let people leave crypto and never come back..

Those who will stay here should stay for “what crypto can really achieve for society, and how we can build tools for a better society using crypto, through which we can also create incentives in the economy not directed by VCs or any institutions but directed by two humans committing to each other”.. commitment and coordination are two core human social problems, and crypto has all the building blocks on which thousands of new markets could be created to make people stronger, more productive, not degens.

I am just locked in to show you what’s possible.. it’s a good time for people who really want to build something great in crypto.. just hang in there if you connect with crypto as something powerful enough to disrupt the old order and build a better new order for human society, or create building blocks for an agentic society to achieve the right coordination and commitments.

Not the way how you see today.

Crypto as counter culture against everything which is happening wrong.

Code is the outlaw.

When you can make code talk privately in an environment where censorship doesn't exist that's when you talk about the real change.

The change which we have been waiting since long.

The change which might make the life of our kids great

The change which demand sacrifices and when we get atleast 1000 people on CT who are ready to grind daily not for money but for purpose

That's the day we will begin to seed plant of change.

It's like a group of monkeys assembling stones to build a bridge to help Rama to conquer ravana.

Rama is crypto here.

Ravana is system design behind traditional Institutions including banks and governments which favour top to bottom value extraction.

Those who will spread the message of crypto in right way are like Hanuman.

Fearless. Strong. Retard. Spiritual. Mystics.

a last word

You don't really need to spend couple of hours just scrolling reactions of different people on unexpected events which gonna keep happening in the world.. if you are wasting your time and energy thinking and talking about these events than you simply wasting a generational opportunity to learn and ship at the speed

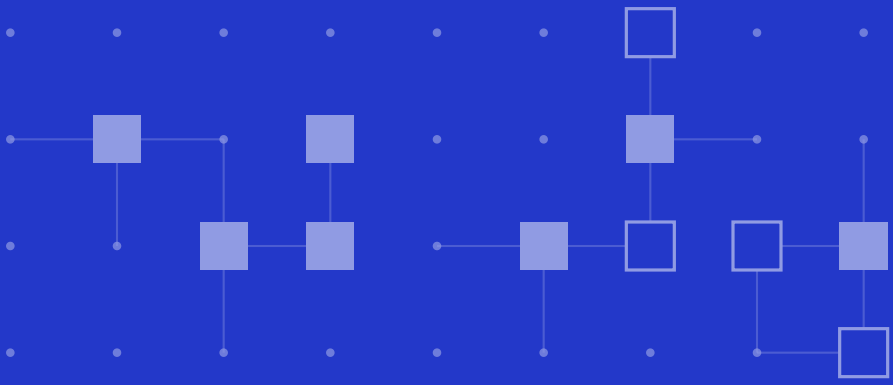
the humanity had never seen in past thousands years. Just think about it, and reflect on your daily activities on the internet.. are you really learning and implementing knowledge on daily basis.. if you are not than it's high time to reinvent yourself.. there is no other way

i started this book with a dream about satoshi, and a man who finally had nothing left to chase. i'll end it with what that freedom made possible for me: the space to think about what crypto is actually for.

it was never only about money. it was about rules we choose, coordination without rulers, privacy as a right, identity we own, reputation we earn, and commitments we keep. money was the first thing we made programmable.

society is next.

and if you have read this far, you are probably one of the people who will build it.



the core objective of the space was never price discovery but building a decentralised society.

hitesh.eth